

TSception: Capturing Temporal Dynamics and Spatial Asymmetry From EEG for Emotion Recognition

Yi Ding^{ID}, *Student Member, IEEE*, Neethu Robinson^{ID}, *Member, IEEE*, Su Zhang^{ID},
Qiuhaio Zeng^{ID}, and Cuntai Guan^{ID}, *Fellow, IEEE*

Abstract—The high temporal resolution and the asymmetric spatial activations are essential attributes of electroencephalogram (EEG) underlying emotional processes in the brain. To learn the temporal dynamics and spatial asymmetry of EEG towards accurate and generalized emotion recognition, we propose TSception, a multi-scale convolutional neural network that can classify emotions from EEG. TSception consists of dynamic temporal, asymmetric spatial, and high-level fusion layers, which learn discriminative representations in the time and channel dimensions simultaneously. The dynamic temporal layer consists of multi-scale 1D convolutional kernels whose lengths are related to the sampling rate of EEG, which learns the dynamic temporal and frequency representations of EEG. The asymmetric spatial layer takes advantage of the asymmetric EEG patterns for emotion, learning the discriminative global and hemisphere representations. The learned spatial representations will be fused by a high-level fusion layer. Using more generalized cross-validation settings, the proposed method is evaluated on two publicly available datasets DEAP and MAHNOB-HCI. The performance of the proposed network is compared with prior reported methods such as SVM, KNN, FBFgMDM, FBTSC, Unsupervised learning, DeepConvNet, ShallowConvNet, and EEGNet. TSception achieves higher classification accuracies and F1 scores than other methods in most of the experiments. The codes are available at: <https://github.com/yi-ding-cs/TSception>

Index Terms—Deep learning, convolutional neural networks, electroencephalography, emotion recognition

1 INTRODUCTION

EMOTIONS are fundamental factors in human beings' daily life [1], affecting decision-making, perception, human interaction, and human intelligence [2]. Emotion recognition plays an important role in Cognitive Behavioural Therapy (CBT) [3], Emotion Regulation Therapy (ERT)/Emotion-Focused Therapy (EFT) [4], [5], [6], and the evaluation of medical treatment [7] for emotion-related mental disorders, such as Generalized Anxiety Disorder (GAD) [8], and Depression [9]. With the potential applications in CBT and EFT, enabling Artificial Intelligence (AI) to identify human emotions has captured more and more interest from researchers recently [1].

Electroencephalography (EEG) is one of the widely used brain imaging technologies, which measures human brain activity directly. Several electrodes are placed on the surface of the human head to collect EEG signals. EEG has high temporal resolution so that it can capture varying brain states at the sub-second level. A Brain-Computer Interface (BCI) system can identify human emotions through EEG,

with the help of machine learning and signal processing techniques [10].

Recently, using EEG-BCI for emotion recognition has gained popularity among researchers [1], [11]. Atkinson *et al.* [12] improved the SVM classifier accuracy for emotion detection by selecting features efficiently, with the accuracy being 73.14%. Zheng *et al.* [13] used a discriminative graph regularized extreme learning machine to investigate stable patterns over time from the differential entropy (DE) features of emotional EEG. Li *et al.* [14] utilized phase-locking value to construct emotion-related brain networks with multiple feature fusion to detect emotions from EEG. Recently, deep learning-based methods have shown promising results in the BCI domain, such as motor imagery classification [15], [16], [17], [18], [19], emotion recognition [20], [21], [22], [23], [24], [25], and mental-task classification [26], [27], [28]. Yang *et al.* [20] designed a hierarchical network structure to perform emotion classification, proposing sub-network nodes to enhance the performance. Li *et al.* [21] constructed EEG into 2D images and proposed a Hierarchical Convolutional Neural Networks (HCNN) to extract the spatial patterns of the EEG. Li *et al.* [22] applied 18 kinds of linear and non-linear features to solve the cross-subject emotion recognition problems, achieving 59.06% and 83.33% on two public datasets. Zhang *et al.* [25] utilized recurrent neural networks (RNN) to learn the temporal-spatial information from the DE features of EEG for emotion recognition. Although many machine learning methods have been proposed for emotion recognition, most of them highly rely on hand-crafted features.

- The authors are with the School of Computer Science and Engineering, Nanyang Technological University, Singapore 639798.
E-mail: {ding.yi, nrobinson, su012, qiuhaio.zeng, ctguan}@ntu.edu.sg.

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(Corresponding Author: Cuntai Guan.)

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TSception：从脑电图捕获时间动态和空间不对称性以进行情绪识别

丁怡， 学生会会员，IEEE，罗尼思·罗宾逊， 会员，IEEE，张苏， 曾秋浩，以及关存泰， Fellow，IEEE

摘要—脑电图（EEG）在脑部情绪过程中的高时间分辨率和不对称空间激活是其重要属性。为了学习 EEG 的时间动态和空间不对称性以实现准确和泛化的情绪识别，我们提出了 TSception，一种可以从 EEG 分类情绪的多尺度卷积神经网络。TSception 由动态时间、不对称空间和高级融合层组成，这些层同时学习时间和通道维度上的判别性表示。动态时间层由与 EEG 采样率相关的多尺度 1D 卷积核组成，学习 EEG 的动态时间和频率表示。不对称空间层利用情绪的不对称 EEG 模式，学习判别性的全局和半球表示。学习到的空间表示将由高级融合层进行融合。在更泛化的交叉验证设置下，该方法在两个公开数据集 DEAP 和 MAHNOB-HCI 上进行了评估。所提出的网络性能与先前报道的方法进行了比较，包括 SVM、KNN、FBFgMDM、FBTSC、无监督学习、DeepConvNet、ShallowConvNet 和 EEGNet。在大多数实验中，TSception 实现了比其他方法更高的分类准确率和 F1 分数。代码可在以下地址获取：<https://github.com/yi-ding-cs/TSception>

索引词—深度学习，卷积神经网络，脑电图，情绪识别



1 引言

情绪是人类日常生活的基础因素[1]，影响决策、感知、人际交往和人类智能[2]。情绪识别在认知行为疗法（CBT）[3]、情绪调节疗法（ERT）/情绪聚焦疗法（EFT）[4][5][6]以及情绪相关精神疾病（如广泛性焦虑症（GAD）[8]和抑郁症[9]）的医学治疗评估[7]中发挥着重要作用。随着在 CBT 和 EFT 中的潜在应用，使人工智能（AI）能够识别人类情绪最近吸引了越来越多研究者的兴趣[1]。脑电图（EEG）是一种广泛使用的脑成像技术，可以直接测量人脑活动。将多个电极放置在人体头部表面以收集脑电图信号。脑电图具有高时间分辨率，能够捕捉亚秒级的不同脑状态。脑机接口（BCI）系统可以通过脑电图识别人类情绪。

借助机器学习和信号处理技术[10]。最近，使用脑电图-脑机接口（EEG-BCI）进行情绪识别在研究人员中越来越受欢迎[1], [11]。Atkinson 等人[12]通过高效选择特征，提高了支持向量机（SVM）分类器在情绪检测中的准确率，准确率达到 73.14%。Zheng 等人[13]使用判别图正则化极限学习机，从情绪脑电图的差分熵（DE）特征中研究随时间稳定的模式。Li 等人[14]利用锁相值构建多特征融合的情绪相关脑网络，从脑电图中检测情绪。最近，基于深度学习的方法在脑机接口（BCI）领域显示出良好的结果，例如运动想象分类[15], [16], [17], [18], [19]、情绪识别[20], [21], [22], [23], [24], [25]和认知任务分类[26], [27], [28]。Yang 等人[20]设计了一种分层网络结构进行情绪分类，提出子网络节点来提高性能。Li 等人[21]将脑电图构建为二维图像，并提出了一种分层卷积神经网络（HCNN）来提取脑电图的时空模式。Li 等人[21] [22] 将 18 种线性和非线性特征应用于解决跨被试情绪识别问题，在两个公开数据集上分别达到了 59.06%和 83.33%的识别率。Zhang 等人[25]利用循环神经网络（RNN）从 EEG 的 DE 特征中学习时空信息以进行情绪识别。尽管已经提出了许多用于情绪识别的机器学习方法，但其中大多数高度依赖于手工特征。

作者来自新加坡南洋理工大学计算机科学与工程学院，邮编 639798。

邮箱：{ding.yi, nrobinson, su012, qiuhao.zeng, ctguan}@ntu.edu.sg。

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With the ability to learn from EEG directly, the convolutional neural networks (CNN) have shown promising results in BCI [18], [29], [30]. Schirrmester *et al.* [15] proposed deep and shallow convolutional neural networks, named DeepConvNet and ShallowConvNet, to process EEG data, combining the feature extraction and classification using a two-stage spatial and temporal input convolution layer. Recently, Lawhern *et al.* [18] proposed EEGNet, which extracts spatial information by the depth-wise convolution kernel whose size is $(n, 1)$. The global spatial dependency can be learned by letting n be the number of channels. All of those networks apply single-scale 1D convolutional kernels along the time and channel dimension to extract temporal and spatial information from EEG.

In order to effectively learn temporal-spatial information from EEG for emotion recognition, several neuro-physiological signatures should be considered. For temporal dimension, EEG signal contains abundant brain activity information in different frequency bands [31]. Due to the non-stationary and dynamic nature of EEG, we hypothesize that a single-sized temporal kernel cannot effectively capture the neural processing underlying emotions that occurs at different time scales and duration. For spatial dimension, especially for emotional processes in the brain, the right and left hemispheres have asymmetric responses to emotions [32]. Hence, we hold the hypothesis that a global spatial kernel has less ability to effectively extract the distinct asymmetric EEG pattern during emotional processes.

To address the above issues, in this paper, we propose TSception, a multi-scale temporal-spatial convolutional neural network to capture temporal dynamics and spatial asymmetry from EEG to classify emotional states. Different from the methods using manually extracted features [21], [22], [25], [33], [34], EEG signals are fed into TSception directly, which makes it an end-to-end deep learning method that needs less domain knowledge about the features. A dynamic temporal layer with different scaled convolutional kernels is proposed to learn richer time-frequency representations from EEG instead of using single-sized temporal CNN kernels [15], [18]. This layer is inspired by the inception block of GoogleNet [35]. Besides the global kernel utilized in [15], [18], we take the brain emotional asymmetry into the kernel design. A hemisphere kernel whose length equals the number of EEG channels located on the right/left hemisphere is proposed to extract the hemisphere asymmetric pattern. The effectiveness of multi-scale convolutional neural networks is preliminarily explored in our previous work [36]. We further propose a high-level fusion layer after asymmetric spatial layer to learn from combined hemisphere-global representations to distinct emotion-class specific information as well as make the network more compact for online usage in the future.

Emotion classification experiments on two publicly available benchmark datasets, a Database for Emotion Analysis using Physiological signals (DEAP) [33], and a multimodal database for affect recognition and implicit tagging (MAHNOB-HCI) [37] were conducted to evaluate the performance of TSception. The generalized cross-validation settings are utilized to avoid potential data leakage and biased evaluation. TSception is compared with several deep and non-deep state-of-the-art methods in the BCI domain, namely

SVM [33], KNN [34], DeepConvNet [15], ShallowConvNet [15], EEGNet [18], Unsupervised learning [38], FBFgMDM [39], and FBTSC [39]. In most of the experiments, the performance of TSception in terms of accuracy and F1 score is higher than the other methods while having a relatively lesser number of network parameters. After statistical analysis, extensive ablation studies are conducted to analyze the contribution of each module in TSception. The saliency map method [45] is utilized to get the most informative part of the EEG data identified by the network. The maps show that the network mainly learns from frontal, temporal, and parietal areas. Frontal, parietal and temporal are commonly known as the functional brain areas related to the emotional processes in the brain [1].

The major contributions of this work can be summarised as:

- We propose TSception, a novel multi-scale temporal-spatial convolutional neural network, for EEG emotion recognition tasks. Several neuro-physiological signatures are involved in the network design. The proposed multi-scale temporal/spatial convolution kernels can capture temporal dynamics and spatial asymmetry from EEG to classify emotions. A high-level fusion layer is proposed to further learn from hemisphere-global representations and to make the network more compact, which can benefit the online usage of TSception in the future.
- Extensive ablation studies and interpretability experiments are conducted to understand the importance of each module in TSception and what it learns using saliency maps.

The PyTorch implementation of TSception is available at <https://github.com/yi-ding-cs/TSception>

The remainder of this article is organized as follows. A summary of related works is introduced in Section 2. In Section 3 the details of TSception are introduced. Section 4 describes the datasets and experiment settings. The result and analysis are given in Section 5. Finally, we discuss the significance of our results in Section 6.

2 MULTI-SCALE CONVOLUTIONAL NEURAL NETWORKS

The detailed instruction of the proposed TSception, a multi-scale convolutional neural network, is presented in this section. EEG data can be treated as 2D time series, whose dimensions are channels (EEG electrodes) and time respectively. The time dimension reflects the brain activity changes from time to time. The spatial dimension can show the brain activation patterns across different functional areas due to the different locations of the electrodes on the brain. EEG signals contain abundant information in different frequency bands [31]. TSception is proposed to identify the most distinct time-frequency-channel specific EEG features corresponding to the emotional states of the user. TSception incorporates specially designed network modules namely, dynamic temporal layer, asymmetric spatial layer, and high-level fusion layer. To extract more discriminative time-frequency representations, multi-scale 1D convolutional kernels are utilized in the dynamic temporal layer to enrich the

能够直接从脑电图（EEG）中学习，卷积神经网络（CNN）在脑机接口（BCI）[18], [29], [30]中已展现出良好的效果。

Schirrmeister 等人[15]提出了深度和浅层卷积神经网络，分别命名为 DeepConvNet 和 ShallowConvNet，用于处理脑电图数据，通过两阶段的空间和时间输入卷积层结合特征提取和分类。最近，Lawhern 等人[18]提出了 EEGNet，该网络通过大小为 $(n; 1b)$ 的深度卷积核提取空间信息。通过将 n 设为通道数，可以学习全局空间依赖性。所有这些网络都在时间和通道维度上应用单尺度 1D 卷积核，从脑电图中提取时间和空间信息。

为了有效地从脑电图（EEG）中学习时空信息以进行情绪识别，应考虑几种神经生理特征。在时间维度上，脑电图信号在不同频段包含了丰富的脑活动信息[31]。由于脑电图的非平稳性和动态特性，我们假设单一尺寸的时间卷积核不能有效地捕捉在不同时间尺度和持续时间下发生的情绪相关神经处理过程。在空间维度上，特别是对于大脑中的情绪过程，左右半球对情绪的响应是非对称的[32]。因此，我们假设全局空间卷积核在情绪过程中提取不同非对称脑电图模式的能力较弱。

为解决上述问题，本文提出 TSception，一种多尺度时-空卷积神经网络，用于从脑电图（EEG）中捕获时序动态和空间不对称性以分类情绪状态。与使用手动提取特征的方法[21]、[22]、[25]、[33]、[34]不同，EEG 信号直接输入 TSception，使其成为一种端到端的深度学习方法，需要较少的关于特征领域知识。我们提出一个具有不同尺度卷积核的动态时序层，以从 EEG 中学习更丰富的时间-频率表示，而不是使用单一尺寸的时序 CNN 核[15]、[18]。该层受 GoogleNet[35]中的 Inception 模块启发。除了[15]、[18]中使用的全局核外，我们将大脑情绪不对称性纳入核设计中。我们提出一个半球的核，其长度等于位于右侧/左侧半球的 EEG 通道数，以提取半球不对称模式。我们之前的工作[36]初步探索了多尺度卷积神经网络的有效性。我们进一步提出在非对称空间层之后添加一个高级融合层，以从半球全局表示中学习区分特定情绪类别的信息，并使网络在未来在线使用时更加紧凑。

为了评估 TSception 的性能，我们在两个公开可用的基准数据集上进行了情绪分类实验，分别是用于生理信号情绪分析的数据库（DEAP）[33]和多模态情感识别与隐式标记数据库（MAHNOB-HCI）[37]。采用广义交叉验证设置以避免潜在的数据泄露和偏差评估。TSception 与 BCI 领域中的几种深度和非深度最先进方法进行了比较。

即 SVM [33]、KNN [34]、DeepConvNet [15]、ShallowConvNet [15]、EEGNet [18]、无监督学习 [38]、FBFgMDM [39] 和 FBTSC [39]。在大多数实验中，TSception 在准确率和 F1 分数方面的性能高于其他方法，同时网络参数数量相对较少。经过统计分析，进行了广泛的消融研究，以分析 TSception 中每个模块的贡献。使用显著性图方法 [45] 获取网络识别的 EEG 数据中最具信息量的部分。这些图显示网络主要从前额叶、颞叶和顶叶区域学习。前额叶、顶叶和颞叶通常被认为是与大脑情绪过程相关的功能脑区 [1]。

这项工作的主要贡献可以总结为：

我们提出了 TSception，一种用于脑电图情绪识别任务的新型多尺度时空间卷积神经网络。网络设计中涉及多种神经生理学特征。所提出的多尺度时/空卷积核能够从脑电图中捕获时序动态和空间不对称性以分类情绪。我们提出了一种高层融合层，用于进一步学习半球全局表示，使网络更加紧凑，这将有助于 TSception 未来在在线场景中的应用。

我们进行了广泛的消融研究和可解释性实验，以理解 TSception 中每个模块的重要性及其通过显著性图学习的内容。TSception 的 PyTorch 实现可在 <https://github.com/yi-ding-cs/TSception> 获取。

本文的其余部分组织如下。相关工作总结在第二节中介绍。第三节介绍了 TSception 的细节。第四节描述了数据集和实验设置。第五节给出了结果和分析。最后，我们在第六节讨论了我们的结果的意义。

多尺度卷积神经网络 本节详细介绍了所提出的 TSception，一种多尺度卷积神经网络。脑电图（EEG）数据可以视为二维时间序列，其维度分别为通道（脑电图电极）和时间。时间维度反映了脑活动随时间的变化。空间维度可以展示由于电极在脑部不同位置而导致的跨不同功能区域的脑激活模式。脑电图信号在不同频段包含丰富的信息 [31]。TSception 被提出用于识别与用户情绪状态相对应的最显著的时间-频率-通道特定脑电图特征。TSception 结合了特别设计的网络模块，即动态时间层、非对称空间层和高级融合层。为了提取更具判别性的时频表示，动态时间层采用了多尺度一维卷积核来丰富

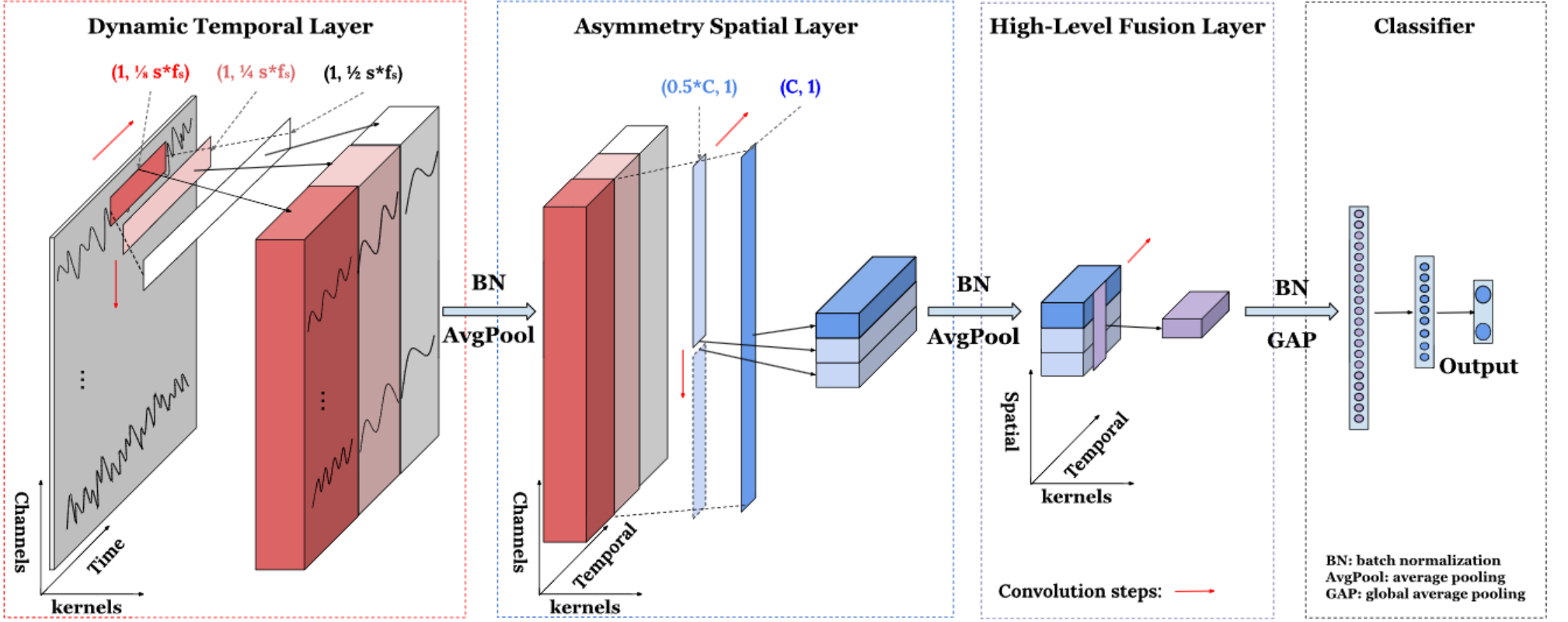


Fig. 1. Structure of TSception. In the figure, f_s is the sampling rate of the EEG signals, C is the number of channels, BN stands for batch normalization, AP is the average pooling operation, and GAP represents global average pooling. TSception has four main parts: the dynamic temporal layer, the asymmetric spatial layer, the high-level fusion layer, and the classifier. The dynamic temporal layer will first learn the dynamic temporal/frequency representations from EEG data channel by channel. After getting the learned representations for each channel, the asymmetric spatial layer will be applied to learn the global spatial representations and the emotional asymmetry pattern using different scale convolutional kernels. To fuse the information from hemisphere and global representations, a high-level fusion layer is utilized. Finally, the fused representation will be passed to the fully connected layers with the softmax as the activation function.

learned time-frequency representations. As for the asymmetric spatial layer, it takes the advantage of neuroscience findings [32] which indicate the brain activities in right and left hemispheres are not symmetrically related to emotions. A hemisphere kernel is proposed to learn the asymmetric representations between two hemispheres. A high-level fusion layer is further proposed to learn from the learned representations of both the hemisphere and global kernels and make the network more compact for real-time usage. The network structure of TSception is shown in Fig. 1. A detailed description of the temporal, spatial, and high-level fusion layers will be discussed in this section.

2.1 Dynamic Temporal Layer

The dynamic temporal layer consists of multi-scale 1D temporal kernels (T kernels). In order to enable the neural network to learn dynamic temporal representations, we set the length of the temporal kernels as the specific ratios of sampling rate f_s of EEG. These ratios are defined as $\alpha^i \in \mathbb{R}$, where i is the level of the dynamic temporal layer. i will vary from 1 to L , if the dynamic temporal layer has L levels. Hence s_T^i , the size of T kernels in i -th level, can be defined as:

$$s_T^i = (1, \alpha^i \cdot f_s), i \in [1, 2, 3]. \quad (1)$$

From the frequency perspective, the length of the T kernel is set as half the sampling rate in EEGNet, allowing for capturing frequency information at 2 Hz and above [18]. Activations related to emotions are observed in Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (> 30 Hz) bands [1]. In this work, we expand the temporal receptive-field, letting $L = 3, i = 1$ to 3, and $\alpha = 0.5$, the ratio coefficients will become [0.5, 0.25, 0.125], learning diversified frequency representations. We hypothesize that the multi-scale temporal kernels can enrich the learned dynamic frequency representations from EEG, providing more emotion-related information.

From the time perspective, multi-scale T kernels can capture long short-term temporal patterns, and learn more diverse representations. The higher level T kernel has a smaller ratio coefficient, which gives a shorter convolutional kernel length and vice versa. The long temporal kernel can learn long-term temporal and low-frequency diverse representations. The short kernel extracts short-term temporal and high-frequency representations. Let X denote EEG input samples. $X = [X^0, X^1, \dots, X^n]$, $X^n \in \mathbb{R}^{c \times l}$, where n is the number of EEG samples, c is the number of channels, l is the length of each sample. The dynamic temporal representations can be generated by parallelly applying the multi-scale temporal kernels on the input EEG samples. After $\text{LeakyReLU}(\cdot)$ activation function, the feature map is further down-sampled by average pooling (AP). The reason for using average pooling is to reduce the effect of the noise as well as the feature dimension since EEG signals are of high dimensions with a low signal-noise ratio. Let $Z_{temporal}^i$ denote the output of the i -th level temporal kernel, $Z_{temporal}^i \in \mathbb{R}^{n \times t \times c \times f_i}$, where n is the number of samples, t is the number of each level's T kernel, c is the number of channels, and f_i is the length of the feature after i -th level convolution operation. $Z_{temporal}^i$ is defined as:

$$Z_{temporal}^i = \text{AP}(\Phi_{L-\text{ReLU}}(\text{Conv1D}(X, s_T^i))), \quad (2)$$

where s_T^i is the T kernel size, X is the input EEG sample array, $\text{Conv1D}(\cdot)$ is the 1D convolution operation with the kernel size being s_T^i , step being (1,1), and $\Phi_{L-\text{ReLU}}(\cdot)$ is the $\text{LeakyReLU}(\cdot)$ activation function.

The output of each level's T kernel will be concatenated along the feature dimension. In order to reduce the internal covariate shift problems in neural networks, we added batch normalization [40] after the dynamic temporal layer. Hence the final output of the dynamic temporal layer, Z_T , $Z_T \in \mathbb{R}^{n \times t \times c \times \sum f_i}$, is defined as:

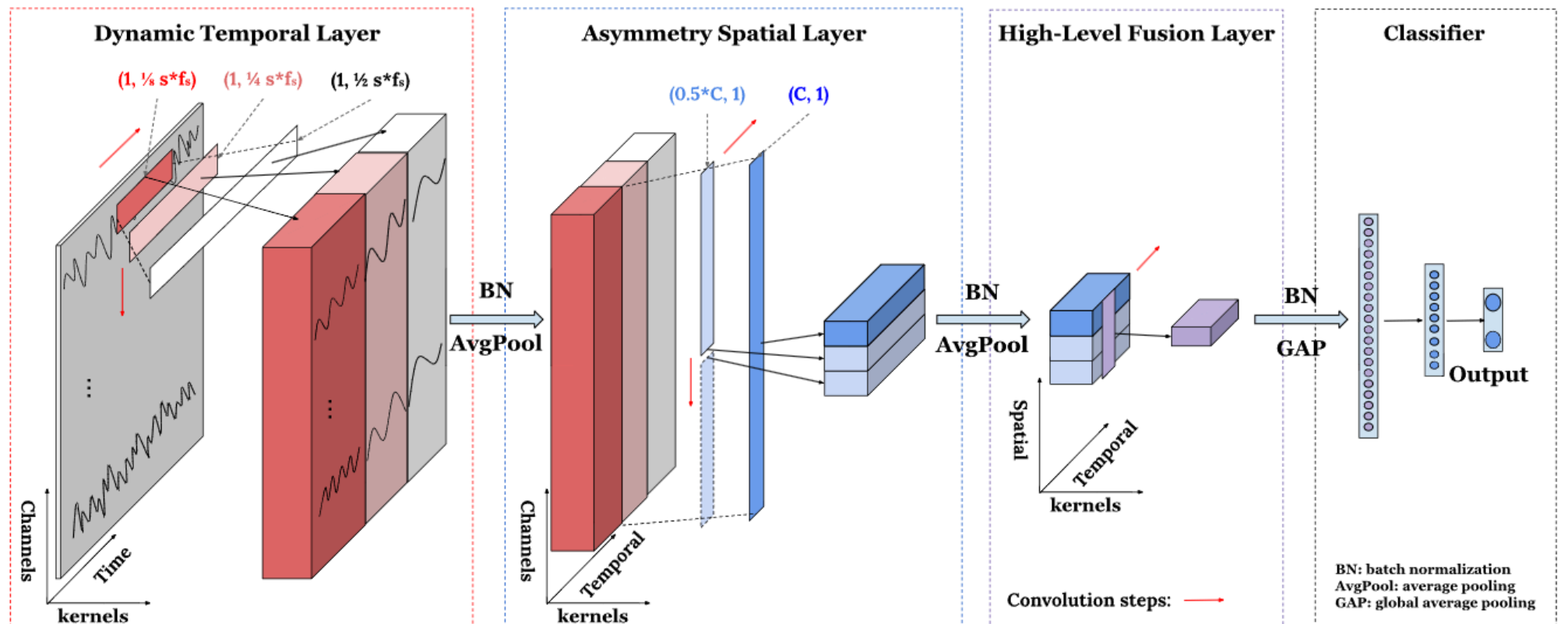


图 1. TSception 的结构。在图中， f 是脑电信号的采样率， C 是通道数，BN 代表批量归一化，AP 是平均池化操作，GAP 表示全局平均池化。TSception 有四个主要部分：动态时间层、非对称空间层、高级融合层和分类器。动态时间层首先从脑电数据中逐通道学习动态时间/频率表示。在得到每个通道的学习表示后，应用非对称空间层，使用不同尺度的卷积核学习全局空间表示和情感非对称模式。为了融合半球和全局表示的信息，使用了一个高级融合层。最后，融合表示将传递给使用 softmax 作为激活函数的全连接层。

学习到的时间-频率表示。对于非对称空间层，它利用了神经科学发现[32]，这些发现表明大脑左右半球的脑活动与情绪没有对称关系。提出了一种半球核来学习两个半球之间的非对称表示。进一步提出了一种高级融合层，从半球核和全局核的学习表示中学习，使网络更加紧凑，便于实时使用。TSception 的网络结构如图 1 所示。时间、空间和高级融合层的详细描述将在本节讨论。

2.1 动态时间层

动态时间层由多尺度 1D 时间核（T 核）组成。为了使神经网络能够学习动态时间表示，我们将时间核的长度设置为 EEG 采样率 f 的特定比率。这些比率定义为 $a \cdot f$ ，其中 a 是动态时间层的级别。如果动态时间层有 L 个级别， a 将从 1 变化到 L 。因此，第 i 级 T 核的大小 s 可以定义为：

$$s = 1; a \cdot f \quad ; i \in [1, 2, 3, \dots] \quad (1)$$

从频率角度来看，EEGNet 中 T 核的长度设置为采样率的一半，以便捕捉 2 Hz 及以上的频率信息[18]。观察到的与情绪相关的激活出现在 Alpha (8-12 Hz)、Beta (12-30 Hz) 和 Gamma (>30 Hz) 频段[1]。在这项工作中，我们扩展了时间感受野，设置 $L=3$ ； $i=1$ 到 3， $a=0.5$ ，比例系数将变为 [0.5, 0.25, 0.125]，学习多样化的频率表示。我们假设多尺度时间核可以丰富从 EEG 学习到的动态频率表示，提供更多与情绪相关的信息。

从时间角度来看，多尺度 T 核能够捕捉长期短期时间模式，并学习更多样化的表示。高级 T 核具有较小的比例系数，这会给出较短的卷积核长度，反之亦然。长时核可以学习长期时间和低频的多样化表示。短核提取短期时间和高频表示。用 XX 表示 EEG 输入样本。 $XX = [XX_1, XX_2, \dots, XX_n]$ ； $XX \in \mathbb{R}$ ，其中 n 是 EEG 样本的数量， c 是通道的数量， l 是每个样本的长度。通过并行地将多尺度时间核应用于输入的 EEG 样本，可以生成动态时间表示。在 LeakyReLU(α) 激活函数之后，特征图通过平均池化（AP）进一步下采样。使用平均池化的原因是减少噪声的影响以及特征维度，因为 EEG 信号具有高维度和低信噪比。 ZZ_{temporal} 表示第 i 级时间卷积核的输出， $ZZ_{\text{temporal}} \in \mathbb{R}$ ，其中 n 是样本数量， t 是每级 T 卷积核的数量， c 是通道数， f 是第 i 级卷积操作后特征的长度。 ZZ_{temporal} 定义为：

$$Z_{\text{temporal}} = \text{AP}(\text{F}(\text{Conv1D}(XX, s))) \quad (2)$$

其中 s 是 T 卷积核的大小， XX 是输入的 EEG 样本数组， $\text{Conv1D}()$ 是步长为 (1,1) 且核大小为 s 的一维卷积操作， $\text{F}()$ 是 LeakyReLU(α) 激活函数。

每级 T 卷积核的输出将沿特征维度连接。为了减少神经网络中的内部协变量偏移问题，我们在动态时间层之后添加了批量归一化 [40]。因此，动态时间层的最终输出 Z ， $Z \in \mathbb{R}$

$$Z_T = f_{bn}([Z_{temporal}^1, \dots, Z_{temporal}^i]), i \in [1, 2, 3], \quad (3)$$

where f_{bn} is the batch normalization operation, and $[\cdot]$ stands for concatenation operation along the feature (f) dimension.

2.2 Asymmetric Spatial Layer

The asymmetric spatial layer has multi-scale 1D convolutional kernels whose sizes are related to the location of the EEG channels. There are two types of spatial kernels: global kernel and hemisphere kernel.

The global kernel has a size of $(c, 1)$, where c is the number of channels. Since the length of the kernel is the same as the channel dimension of the input EEG segment, it can learn the global spatial information.

In this work, we further combine the frontal area of brain emotional asymmetry [41] into the kernel design. The hemisphere kernel is used to extract the relations between the left and right hemispheres by sharing the convolutional kernels. The size of the hemisphere kernel is $(0.5 \cdot c, 1)$, and the step is $(0.5 \cdot c, 1)$, where c is the total number of channels. The hemisphere kernel is shared by two hemispheres without overlapping so that the asymmetric pattern can be extracted. The size of the spatial kernel s_S^j can be defined as:

$$s_S^j = (\delta^j \cdot c, 1), j \in [0, 1], \quad (4)$$

where $\delta = 0.5$ is the coefficient to control the ratio between the spatial kernel length and the total number of channels.

Let $Z_{spatial}^j$ denote the output of the j -th type spatial kernel, $Z_{spatial}^j \in \mathbb{R}^{n \times s \times c_j \times f}$, where n is the number of samples, s is the number of each type S kernel, c_j is the number of channels after j -th spatial convolution, and f is the length of the feature after each spatial convolution operation. $Z_{spatial}^j$ is defined as:

$$Z_{spatial}^j = AP(\Phi_{L-ReLU}(Conv1D(Z_T, s_S^j))), \quad (5)$$

where s_S^j is the S kernel size, Z_T is the output of dynamic temporal layer, $Conv1D(\cdot)$ is the 1D convolution operation with the kernel size being s_S^j , the step being $(1, 1)$ for the global kernels and $(0.5 \cdot c, 1)$ for the hemisphere kernels, and $\Phi_{L-ReLU}(\cdot)$ is the LeakyReLU($\cdot$) activation function.

In order to apply hemisphere kernels, the sequence of channels in the input EEG samples should be arranged in a particular way. The order of the channels should be $[channel_{left}, channel_{right}]$, where the $channel_{left}$ are the channels located in the left hemisphere, the $channel_{right}$ are the ones on the right hemisphere. The order for channels on each hemisphere should also be rearranged to make each kernel weight shared between pairs of symmetrically located electrodes on two hemispheres because the step of the hemisphere kernel is also $(0.5 \cdot c, 1)$. Fig. 2 shows the electrode locations of DEAP dataset. The final output of the asymmetric spatial layer, $Z_S, Z_S \in \mathbb{R}^{n \times s \times \sum c_j \times f}$ is defined as:

$$Z_S = f_{bn}([Z_{spatial}^0, \dots, Z_{spatial}^j]), j \in [0, 1], \quad (6)$$

where f_{bn} is the batch normalization operation, and $[\cdot]$ stands for concatenation operation along the channel (c) dimension. The output of hemisphere kernel have a length of two in the spatial dimension, which refers to two hemispheres respectively. The output of global kernel is only a vector whose

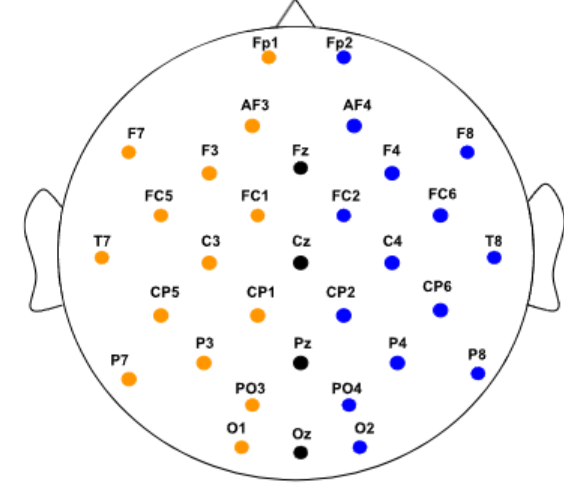


Fig. 2. The location map of 32 channels cap. The electrodes can be divided into 3 groups: electrodes on the left hemisphere (in orange), electrodes on the right hemisphere (in blue), and the electrodes on the central line (in black). For the electrodes located on the central line of the head, Fz, Cz, Pz, and Oz, which can not be paired on the left and right hemispheres, we further removed them to let TSception learn the asymmetric pattern of left and right hemispheres better.

length in the channel dimension is one. After concatenation, the channel dimension is $\sum c_j = 3$.

2.3 High-Level Fusion Layer

In order to learn high-level spatial representations by fusing the learned information from global and hemispheres, a high-level fusion layer is further proposed. Given the output of asymmetric spatial layer, $Z_S \in \mathbb{R}^{n \times s \times 3 \times f}$, a 1D convolutional layer whose kernel size is $(3, 1)$ is utilized to fuse the information along the spatial dimension. After LeakyReLU($\cdot$), average pooling, and batch normalization, a global average pooling layer (GAP) is added to overcome over-fitting and reduce the model size. The final learned global-hemisphere fusion representations will be generated by:

$$Z_{fusion} = GAP(f_{bn}(AP(\Phi_{L-ReLU}(Conv1D(Z_S, (3, 1)))))). \quad (7)$$

Finally, the latent representation of Z_{fusion} will be fed into fully connected layers. The final output layer is activated by the softmax function, $\Phi_{softmax}(\cdot)$. Hence the final output can be calculated by:

$$\text{Output} = \Phi_{softmax}(W' \Phi_{dp}(\Phi_{ReLU}(W(\Gamma(Z_{fusion}))) + b)) + b'), \quad (8)$$

where the $\Gamma(\cdot)$ is the squeeze operation, W and W' are the trainable weight matrix, b and b' are the bias terms.

The proposed TSception is summarised in Algorithm 1.

The structure of the proposed TSception is shown in Table 2.

3 EXPERIMENTS

3.1 Datasets

To evaluate the proposed TSception, we conducted several experiments on two publicly available benchmark datasets, a Database for Emotion Analysis using Physiological signals (DEAP)¹ [33], and a multimodal database for affect recognition and implicit tagging (MAHNOB-HCI)² [37]. Table 1 summarizes the related information of the two datasets

1. <http://www.eecs.qmul.ac.uk/mmv/datasets/deap/index.html>

2. <https://mahnob-db.eu/hci-tagging/>

$$Z = f(\frac{1}{2}ZZ_{\text{temporal}}; ::; ZZ_{\text{temporal}}); i \in \frac{1}{2}1; 2; 3; \quad (3)$$

其中 f 表示批量归一化操作, $\frac{1}{2}$ 表示沿特征 (f) 维度进行的拼接操作。

2.2 非对称空间层

非对称空间层具有多尺度 1D 卷积核, 其大小与 EEG 通道的位置相关。空间核有两种类型: 全局核和半球核。

全局核的大小为 $(\delta c; 1P$, 其中 c 是通道数。由于核的长度与输入 EEG 段的通道维度相同, 它可以学习全局空间信息。

在这项工作中, 我们将大脑情感不对称的额叶区域[41]进一步结合到核设计中。半球核通过共享卷积核来提取左右半球之间的关系。半球核的大小为 $(0.5 * c; 1)$, 步长为 $(0.5 * c; 1)$, 其中 c 是总通道数。两个半球共享半球核且不重叠, 以便提取不对称模式。空间核 s 的大小可以定义为:

$$s = d * c; 1; j \in [0; 1]; \quad (4)$$

其中 $d = 0.5$ 是控制空间核长度与总通道数之间比例的系数。

ZZ 空间表示第 j 种类型空间核的输出, $ZZ \text{ 空间} \in R$, 其中 n 是样本数量, s 是每种类型 S 核的数量, c 是第 j 次空间卷积后的通道数, f 是每次空间卷积操作后的特征长度。 ZZ 空间定义为:

$$Z \text{ 空间} = AP(F(\text{Conv1D}(ZZ; s))) \quad (5)$$

其中 s 是 S 核的大小, ZZ 是动态时间层的输出, $\text{Conv1D}()$ 是核大小为 s 的一维卷积操作, 全局核的步长为 $(1,1)$, 半球的核的步长为 $(0.5 c; 1)$, F 是 $\text{LeakyReLU}()$ 激活函数。

为了应用半球核, 输入的 EEG 样本中通道的序列需要以特定方式排列。通道的顺序应为 $\frac{1}{2}\text{channel}; \text{channel}$, 其中 channel 表示位于左半球的通道, channel 表示位于右半球的通道。每个半球的通道顺序也需要重新排列, 以便每个核权重在两个半球上对称位置的电极对之间共享, 因为半球核的步长也是 $(0.5 c; 1)$ 。图 2 显示了 DEAP 数据集的电极位置。非对称空间层的最终输出 ZZ , $ZZ \in R$

$$P_{c \times f} \text{ 定义为:}$$

$$Z \frac{1}{2} f \frac{1}{2} ZZ_{\text{spatial}}; ::; ZZ_{\text{spatial}}; j \in \frac{1}{2}0; 1; \quad (6)$$

f 表示批量归一化操作, $\frac{1}{2} \frac{1}{2}$ 表示沿通道 (c) 维度进行的拼接操作。半球的核输出在空间维度上有两个长度, 分别指代两个半球。

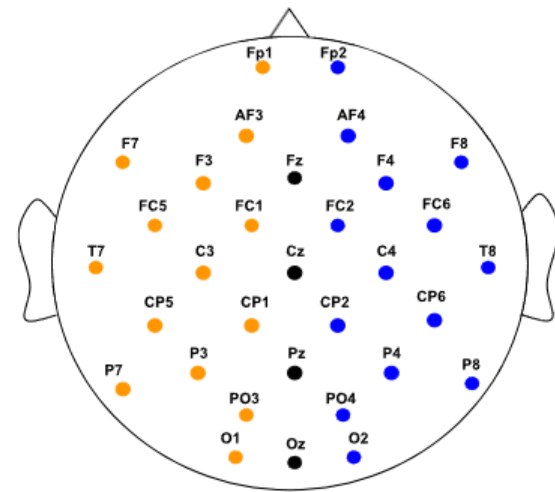


图 2. 32 个电极的位置图。电极可以分为三组: 左侧半球的电极 (橙色)、右侧半球的电极 (蓝色) 和中线上的电极 (黑色)。对于位于头部中线的电极, Fz 、 Cz 、 Pz 和 Oz , 这些电极在左右半球上无法配对, 我们进一步将它们移除, 以便 TSception 更好地学习左右半球的不对称模式。

全局核的输出只是一个长度在通道维度上为 1 的向量。拼接后, 通道维度为 $c=3$ 。

2.3 高级融合层

为了通过融合来自全局和半球的已学习信息来学习高级空间表示, 进一步提出了一个高级融合层。给定非对称空间层的输出 $ZZ \in R$, 使用一个内核大小为 $\delta 3; 1P$ 的一维卷积层来融合空间维度的信息。经过 $\text{LeakyReLU}()$ 、平均池化和批量归一化后, 添加了一个全局平均池化层 (GAP) 以克服过拟合并减小模型大小。最终通过以下方式生成全局-半球融合表示:

$$ZZ \frac{1}{4} \text{GAP} \delta f \delta AP \delta F \delta \text{Conv1D} \delta ZZ; \delta 3; 1P \delta \delta \delta \delta \delta \delta \delta \quad (7)$$

最后, ZZ 的潜在表示将被输入到全连接层。最终输出层由 softmax 函数 $F \delta P$ 激活。因此, 最终输出可以通过以下方式计算:

$$\text{Output} \frac{1}{4} F \delta WW \delta F \delta WW \delta G \delta ZZ \delta P \delta b \delta P \delta b \delta P; \quad (8)$$

$G \delta P$ 是挤压操作, WW 和 W 是可训练的权重矩阵, b 和 b 是偏置项。

提出的 TSception 总结在算法 1 中。提出的 TSception 的结构如表 2 所示。

3 实验

3.1 数据集

为了评估所提出的 TSception 模型, 我们在两个公开可用的基准数据集上进行了多项实验, 分别是用于生理信号情绪分析的数据库 (DEAP) [33] 和多模态情绪识别与隐式标记数据库 (MAHNOB-HCI) [37]。表 1 总结了这两个数据集的相关信息

TABLE 1
Summary of Related Information of the Datasets
Used in the Experiments

Factor	DEAP	MAHNOB-HCI
Subjects	32	27
Stimuli	Music videos	Emotional videos
Trials/subject	40	20
Trial duration	1 min	35-117s
EEG channels	32	32
Sampling rate	512Hz	256Hz
Label	V/A	V/A

V: valence; A: arousal.

used in our experiments. Arousal and valence dimensions on both datasets were utilized as reported in [39].

DEAP is a multi-modal human affective states dataset, including EEG, facial expressions, and galvanic skin response (GSR). There are 32 subjects watching music video clips while their EEG, facial expression, and GSR are recorded. Each of the subjects participates in 40 trials in total. The duration of each trial is 1 minute with a 3 seconds pre-trial baseline. After each trial, the subject will be given a questionnaire to provide their own emotional state in arousal, valence, dominance, and liking with each dimension having 9 discrete levels. The EEG is collected using 32 channels device, with the sampling rate being 512Hz.

Algorithm 1. TSception

Input: EEG data $X^n \in \mathbb{R}^{c \times l}$; ground truth label y ;
Output: $\hat{y}$, the prediction of TSception

- 1 Initialization;
- 2 **do in sequential**
- 3 # get the output of the dynamic temporal layer
- 4 **for** $i \leftarrow 1$ **to** 3 **do**
- 5 get i -th temporal kernel size by Eq. 1;
- 6 get $z_{temporal}^i$ by Eq. 2 using X^n as input;
- 7 **end**
- 8 get Z_T by Eq. 3;
- 9 # get the output of the asymmetric spatial layer
- 10 **for** $j \leftarrow 0$ **to** 1 **do**
- 11 get j -th temporal kernel size by Eq. 4;
- 12 get $z_{spatial}^j$ by Eq. 2 using Z_T as input;
- 13 **end**
- 14 get Z_S by Eq. 6;
- 15 # get the output of the high-level fusion layer
- 16 get Z_{fusion} by Eq. 7;
- 17 **end**
- 18 get $\hat{y}$ using Eq. 8;

Return $\hat{y}$

MAHNOB-HCI [37] is another multi-modal dataset similar to the DEAP dataset. There are 30 subjects watching movie clips while their facial expression, audio signals, eye gaze data, EEG signal, and other physiological signals are recorded. Note that Subject 12, 15, and 26 failed to finish the data collection, therefore, the remaining 27 out of 30 subjects were used in this work. The movie clips are between 35 and 117 seconds long. The EEG signals are acquired from 32 electrodes on the 10-20 international system. The sampling frequency is 256 Hz. For each trial, four integers ranged

from 1 to 9 and self-reported by the subjects are used to label the valence, arousal, dominance, and emotional keywords, respectively.

3.2 Pre-Processing

For DEAP, the 3 seconds pre-trial baseline was removed for each trial. Then the data was down-sampled from 512Hz to 128Hz, after which the electrooculogram (EOG) was removed with a blind source separation method as [33]. To remove the low and high-frequency noise, a band-pass filter from 4.0-45Hz was applied to the original EEG as [33]. Finally, the EEG channels were averaged to the common reference. The class label for each dimension is from 1 to 9, hence 5 was selected as a threshold to project the 9 discrete values into low and high classes in each dimension as [33], [39]. In line with [39], only arousal and valence dimensions are used in this study. The deep neural networks have a higher number of trainable parameters hence to optimally learn emotion state representations in EEG a large number of labelled data samples are required. However, as listed in Table 1, the number of trials is very small in the selected datasets. To overcome this challenge, a data augmentation step by splitting each trial into smaller non-overlapping 4s segments was applied. The segments were then used to train the deep neural network.

For MAHNOB-HCI, the pre-processing was much the same as that for the DEAP dataset except for the following. First, the 30 seconds pre-trial and post-trial baselines were removed for each trial, so that the remaining corresponds to the event of emotion elicitation [37]. Second, to remove the low-high frequency noise, a band-pass filter from 0.3-45Hz was applied to the original EEG as [36]. Note that the delta band 0.3-4Hz is included since it also contributes to an individual's affective state [42], [43].

3.3 Performance Evaluation Metrics

The first type of metric is accuracy. It is one of the most commonly used evaluation metrics in classification problems [36]. It is the ratio of the correctly predicted samples and the total number of the samples. For binary classification problems, the accuracy can also be defined as:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}, \quad (9)$$

where TP is the true positive, TN is the true negative, and FP is the false positive, and FN is the false negative.

Accuracy can measure how precise the prediction is for the class-balanced dataset. However, after the pre-processing of the labels mentioned in the pre-processing section, the labels become imbalanced. To better evaluate the performance of a classifier on class-imbalanced datasets, the F1 score is added as [33], [38]. It combines the precision and recall of the classifier, and it is defined as the harmonic mean of the classifier's precision and recall. F1 is defined by:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}, \quad (10)$$

where TP is the true positive, TN is the true negative, and FP is the false positive, and FN is the false negative.

表 1 数据集相关信息的汇总

用于实验		
因子	DEAP	MAHNOB-HCI
受试者	32	27
刺激	音乐视频	情绪视频
试验/被试	40	20
试验时长 1 分钟		35-117 秒
脑电图通道	32	32
采样率 512Hz		256Hz
标签	V/A	V/A

V: 效价; A: 唤醒度。

在我们的实验中使用了这些数据集。两个数据集的唤醒度和效价维度如[39]中所述被使用。

DEAP 是一个多模态的人类情感状态数据集，包括脑电图（EEG）、面部表情和皮肤电反应（GSR）。共有 32 名受试者观看音乐视频片段，同时记录他们的 EEG、面部表情和 GSR。每位受试者总共参与 40 次试验。每次试验的持续时间为 1 分钟，并有 3 秒的预试验基线。每次试验后，受试者将填写问卷，提供他们在唤醒度、效价、支配力和喜爱度四个维度上的情感状态，每个维度有 9 个离散级别。EEG 数据通过 32 通道的设备采集，采样率为 512Hz。

算法 1. TSception

输入：EEG 数据 $XX \in \mathbb{R}^2$ ；真实标签 y ；输出：TSception 的预测结果 b ，初始化

2 按顺序执行

3 # 获取动态时间层的输出

4 for i 1 to 3 do

5 通过公式 1 获取第 i 个时间卷积核的大小；

6 使用 XX 作为输入，通过公式 2 获取 $ztemporal$ ；

7 结束

8 通过公式 3 获取 Z ；

9 # 获取非对称空间层的输出

10 for j 0 to 1 do

11 通过公式 4 获取第 j 个时间卷积核大小；

12 使用 Zas 输入，通过公式 2 获取 $zspatial$ ；

13 结束

14 通过公式 6 获取 Z ；

15 # 获取高层融合层的输出

16 通过公式 7 获取 Z ；

17 结束

18 使用公式 8 获取；返回

MAHNOB-HCI [37] 是另一个与 DEAP 数据集类似的多模态数据集。有 30 名受试者观看电影片段，同时记录了他们的面部表情、音频信号、眼动数据、脑电图信号和其他生理信号。请注意，受试者 12、15 和 26 没有完成数据收集，因此本研究使用了剩余的 27 名受试者中的 30 名。电影片段的长度在 35 到 117 秒之间。脑电图信号是从 10-20 国际系统的 32 个电极上采集的。采样频率为 256 Hz。对于每次试验

使用从 1 到 9 的四个整数，由受试者自我报告，分别用于标记效价、唤醒度、支配性和情绪关键词。

3.2 预处理

对于 DEAP 数据集，每个试次的 3 秒预试基线数据被移除。随后，数据从 512Hz 下采样至 128Hz，并使用盲源分离方法[33]去除眼电图（EOG）。为去除低频和高频噪声，根据[33]对原始脑电图应用了 4.045Hz 的带通滤波器。最后，脑电通道被平均到公共参考点。每个维度的类别标签从 1 到 9，因此选择 5 作为阈值，将 9 个离散值投影到每个维度的高低类别中[33][39]。与[39]一致，本研究仅使用唤醒度和效价维度。深度神经网络具有更多的可训练参数，因此为了在脑电图中最优地学习情绪状态表示，需要大量的标记数据样本。然而，如表 1 所示，所选数据集中的试次数量非常小。为克服这一挑战，采用了数据增强步骤，将每个试次分割成更小的非重叠 4 秒片段。这些片段随后用于训练深度神经网络。

对于 MAHNOB-HCI，预处理过程与 DEAP 数据集基本相同，但有以下几点不同。首先，每个试验中移除了 30 秒的试验前和试验后基线，以便剩余部分对应于情绪唤起事件[37]。其次，为了去除低频-高频噪声，对原始脑电图应用了 0.3-45Hz 的带通滤波器，如[36]所述。需要注意的是，0.3-4Hz 的 delta 波段也包括在内，因为它也对个体的情感状态有贡献[42]，[43]。

3.3 性能评估指标

第一种指标是准确率。它是分类问题中最常用的评估指标之一[36]。它是正确预测样本数与样本总数的比值。对于二元分类问题，准确率也可以定义为：

准确率 =

$$\frac{TP \text{ } p \text{ } TN}{TP \text{ } p \text{ } FP \text{ } p \text{ } TN \text{ } p \text{ } FN} ;$$

(9)

其中，TP 代表真阳性，TN 代表真阴性，FP 代表假阳性，FN 代表假阴性。

准确率可以衡量在类别平衡数据集上预测的精确度。然而，在预处理部分提到的标签经过预处理后，标签变得不平衡。为了更好地评估分类器在类别不平衡数据集上的性能，添加了 F1 分数[33]，[38]。它结合了分类器的精确率和召回率，定义为分类器精确率和召回率的调和平均值。F1 分数定义为：

$$F1 \text{ } \frac{1}{4} 2$$

$$\frac{\text{精确率} \text{ } 召回率}{\text{精确率} \text{ } p \text{ } 召回率} \text{ } \frac{1}{4} \text{ } \frac{\text{真阳性}}{TP \text{ } p \text{ } (FP \text{ } p \text{ } FN)} ;$$

(10)

其中，TP 是真阳性，TN 是真阴性，FP 是假阳性，FN 是假阴性。

TABLE 2
Structure of the Proposed TSception

Model structure		Layers	Input	Output
Block1	3 branches (in parallel)	Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 64)	(-1, 1, 28, 512)	(-1, 15, 28, 56)
		Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 32)	(-1, 1, 28, 512)	(-1, 15, 28, 60)
		Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 16)	(-1, 1, 28, 512)	(-1, 15, 28, 62)
		Concatenate, BN		(-1, 15, 28, 178)
Block2	2 branches (in parallel)	Conv2d, LK-ReLU, AP((1,2)) Kernel=15@(28, 1)	(-1, 15, 28, 178)	(-1, 15, 1, 89)
		Conv2d, LK-ReLU, AP((1,2)) Kernel=15@(14, 1), Stride=(14, 1)	(-1, 15, 28, 178)	(-1, 15, 2, 89)
		Concatenate, BN		(-1, 15, 3, 89)
Block3		Conv2d, LK-ReLU, AP((1,4)), BN, GAP Kernel=15@(3, 1)	(-1, 15, 3, 89)	(-1, 15, 1)
		Flatten	(-1, 15, 1)	(-1, 15,)
Fully connected layers		Linear(32), ReLU	(-1, 15,)	(-1, 32,)
		dropout(0.5)	(-1, 32,)	(-1, 32,)
		Linear(2)	(-1, 32,)	(-1, 2,)
		softmax	(-1, 2,)	(-1, 2,)

LK-ReLU is the Leaky-ReLU activation function. AP is the average pooling operation. BN stands for batch normalization. GAP is the global average pooling. '-1' in the tensor size stands for the number of samples within one mini-batch. The strides of CNNs are (1, 1) if not specified, and the one for pooling layers is the same as the pooling step.

3.4 Experiment Settings

There are two types of experiment settings in this paper: I) trial-wise 10-fold cross-validation and II) leave-one-trial-out cross-validation. Each of them is introduced in the following paragraphs.

In the first experiment setting, we split each trial into 4's non-overlapping segments, also know as cropped experiments [15], and a trial-wise 10-fold cross-validation is utilized for each subject to prevent potential data leakage issues. The reason for doing cropped experiments is that the predictions of shorter segments are preferred than the trial-wise predictions that are evaluated in [33], [38], [39] for an efficient real-time BCI system. Besides, a decoding model with a good generalization capability is needed for the real-world situation where the testing data is unseen to the model. In each trial, the subject was asked to watch or hear a certain stimulus that is supposed to evoke a certain type of emotion. Because emotion is one of the continuous cognitive processes in the brain, the data segments within a single trial are highly correlated. Hence, randomly shuffle the segments among trials before the training-testing split of the data could make the adjacent segments be in training and testing data, which will give high classification results. But the accuracy will drop when the highly correlated segments are never seen by the model in the real-world situation. To get the more generalized evaluation, the 10 folds are split among trials, which will make sure the adjacent segments in one trial will not appear in both training and testing data. In each step of 10-fold cross-validation, one fold is selected as testing data, the rest 9 folds are utilized as training data. Among the 9 training folds, the data is randomly divided into 80% training data and 20% validation data. During the training process, we train the network on training data for 500 epochs and evaluate the network on validation data in each epoch. The model with the highest accuracy on validation data among those 500 epochs is saved and tested on the testing data. The above process is repeated 10 times for each subject till each fold has been the testing fold once. In each

fold, the test data remains completely unseen in all stages of training and validation. The mean accuracy and F1 score of all subjects are reported as the final results.

In the second experiment setting, a leave-one-trial-out cross-validation is adopted for each subject to further compare our methods with the recently proposed methods in [39] and [38]. In each cross-validation step, one trial is selected as testing data and the rest are selected as training data. For each step of the leave-one-trial-out cross-validation, the training data is also split into 80% training and 20% validation data. The process is repeated till every trial is selected as testing data once for each subject. The average accuracy and F1 score of all subjects are reported as the final evaluation criterion as [38]. The features extracted from the entire trial's data are utilized as one input sample to the classifier in [33], [38], [39]. To compare our deep learning methods trained from segmented EEG data with those papers, a voting mechanism is utilized for the segment predictions in each testing trial as:

$$\hat{y}_t = \begin{cases} 0 & n_{y=0}^{\wedge} > n_{y=1}^{\wedge} \\ 1 & n_{y=1}^{\wedge} \geq n_{y=0}^{\wedge} \end{cases} \quad (11)$$

where $\hat{y}_t$ is the prediction of one testing trial, n is the number of the predictions of the segments in each trial under each condition indicated in the sub-script.

3.5 Implementation Details

The code is implemented using the PyTorch library, the source code can be found via this link³.

The ratio coefficients of T kernel length are [0.5, 0.25, 0.125] for DEAP. The sampling rate of the data in DEAP is 128Hz, hence, the temporal kernel lengths are 64, 32, and 16 according to Eq. 1. When training TSception on MAHNOB-HCI, we found that using [0.25, 0.125, 0.0625] as the ratio

3. <https://github.com/yi-ding-cs/TSception>

表 2 TSception 结构

Model structure		Layers	Input	Output
Block1	3 branches (in parallel)	Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 64)	(-1, 1, 28, 512)	(-1, 15, 28, 56)
		Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 32)	(-1, 1, 28, 512)	(-1, 15, 28, 60)
		Conv2d, LK-ReLU, AP((1,8)) Kernel=15@(1, 16)	(-1, 1, 28, 512)	(-1, 15, 28, 62)
		Concatenate, BN		(-1, 15, 28, 178)
Block2	2 branches (in parallel)	Conv2d, LK-ReLU, AP((1,2)) Kernel=15@(28, 1)	(-1, 15, 28, 178)	(-1, 15, 1, 89)
		Conv2d, LK-ReLU, AP((1,2)) Kernel=15@(14, 1), Stride=(14, 1)	(-1, 15, 28, 178)	(-1, 15, 2, 89)
		Concatenate, BN		(-1, 15, 3, 89)
Block3		Conv2d, LK-ReLU, AP((1,4)), BN, GAP Kernel=15@(3, 1)	(-1, 15, 3, 89)	(-1, 15, 1)
		Flatten	(-1, 15, 1)	(-1, 15,)
Fully connected layers		Linear(32), ReLU	(-1, 15,)	(-1, 32,)
		dropout(0.5)	(-1, 32,)	(-1, 32,)
		Linear(2)	(-1, 32,)	(-1, 2,)
		softmax	(-1, 2,)	(-1, 2,)

LK-ReLU 是 Leaky-ReLU 激活函数。AP 是平均池化操作。BN 代表批量归一化。GAP 是全局平均池化。张量大小中的'-1'表示一个 mini-batch 内的样本数量。如果未指定，CNN 的步长为(1, 1)，池化层的步长与池化步长相同。

3.4 实验设置

本文中有两种实验设置：I) 试次级别的 10 折交叉验证和 II) 留一
试次交叉验证。它们分别在下文介绍。
在第一个实验设置中，我们将每个试验分成 4 个非重叠的片段，也
称为裁剪实验[15]，并对每个受试者采用试验级别的 10 折交叉验
证，以防止潜在的数据泄露问题。进行裁剪实验的原因是，较短片
段的预测比在[33]、[38]、[39]中评估的试验级别预测更受青睐，这
对于高效的实时 BCI 系统更为有效。此外，对于测试数据对模型来
说是未知的现实情况，需要一个具有良好泛化能力的解码模型。在
每个试验中，受试者被要求观看或聆听某种刺激，该刺激应引发某
种类型的情绪。由于情绪是大脑中的一种连续认知过程，单个试验
内的数据片段高度相关。因此，在数据训练-测试分割之前，随机打
乱试验之间的片段，可以使相邻片段分别出现在训练数据和测试数
据中，这将获得较高的分类结果。但当高度相关的片段在现实情况
下从未被模型看到时，准确率会下降。为了获得更通用的评估，将
10 折数据分配到试验中，以确保同一试验中的相邻片段不会同时出
现在训练和测试数据中。在 10 折交叉验证的每一步中，选择 1 折作
为测试数据，其余 9 折用作训练数据。在 9 折训练数据中，数据随
机分为 80%的训练数据和 20%的验证数据。在训练过程中，我们在
训练数据上训练网络 500 个 epoch，并在每个 epoch 在验证数据
上评估网络。在 500 个 epoch 中，保存验证数据上准确率最高的
模型，并在测试数据上测试该模型。上述过程对每个受试者重复 10
次，直到每个折都作为测试折一次

在每个折中，测试数据在训练和验证的所有阶段都完全未见过。所
有受试者的平均准确率和 F1 分数作为最终结果报告。
在第二个实验设置中，对每个受试者采用留一试验交叉验证，以进
一步将我们的方法与最近提出的[39]和[38]中的方法进行比较。在每
个交叉验证步骤中，选择一个试验作为测试数据，其余的作为训练
数据。对于每个留一试验交叉验证步骤，训练数据也分为 80%训练
数据和 20%验证数据。重复此过程，直到每个受试者的每个试验都
作为一次测试数据被选中。所有受试者的平均准确率和 F1 分数被报
告为最终评估标准[38]。从整个试验数据中提取的特征被用作[33]、
[38]、[39]中分类器的一个输入样本。为了将我们从分割的脑电图数
据中训练的深度学习方法与这些论文进行比较，在每次测试试验的
片段预测中采用投票机制，如下所示：

$$y = \begin{cases} 0, & \text{if } n_b > n_t \\ 1, & \text{if } n_b = n_t \end{cases} \tag{11}$$

其中 y 是单个测试试验的预测，n 是每个试验在每个下标条件下的片
段预测数量。

3.5 实现细节

代码使用 PyTorch 库实现，源代码可通过此链接获取。
DEAP 中 T 核长度的比例系数为[0.5, 0.25, 0.125]。DEAP 中数据
的采样率为 128Hz，因此根据公式 1，时间核长度分别为 64、32 和
16。在 MAHNOBHCI 上训练 TSception 时，我们发现将[0.25,
0.125, 0.0625]作为比例系数

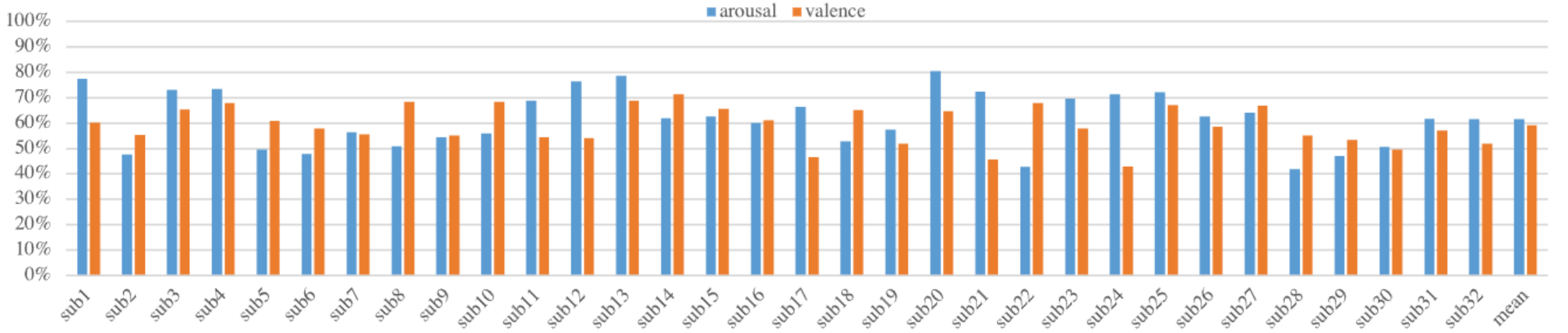


Fig. 3. Mean accuracy of each subject for arousal and valence on DEAP using TSception.

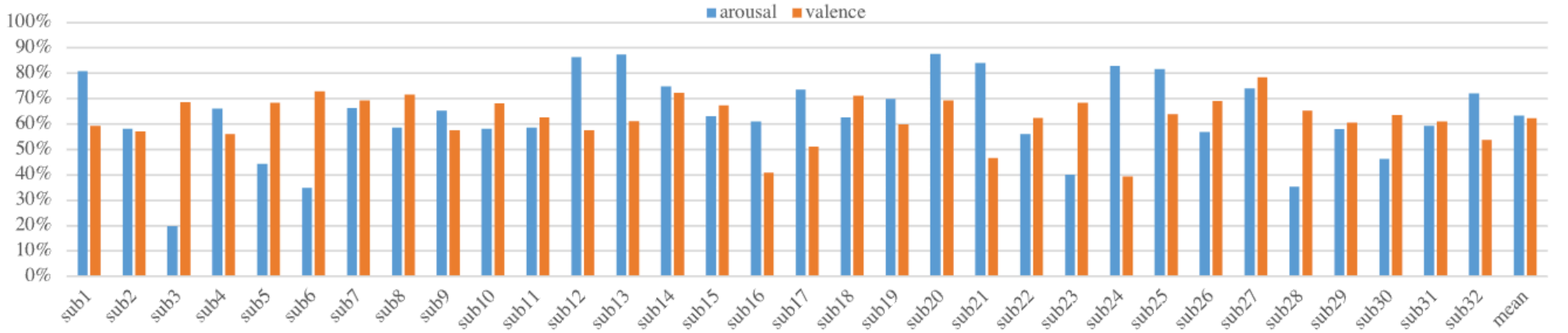


Fig. 4. Mean F1 score of each subject for arousal and valence on DEAP using TSception.

TABLE 3
Trial-Wise 10-Fold Cross-Validation Classification Results of SVM, KNN, EEGNet, ShallowConvNet, DeepConvNet, and TSception on DEAP

Method	Arousal				Valence				Parameters
	ACC	std	F1	std	ACC	std	F1	std	
SVM	60.37%	12.25%	57.33%	26.61%	55.19% **	6.97%	57.87%**	11.36%	NA
KNN	59.48%	12.34%	57.49%	24.96%	53.03% **	9.14%	55.12%**	16.27%	NA
EEGNet	58.29%	8.60%	60.60%	15.20%	54.56% **	8.14%	57.61%**	10.42%	2,162
SCN	61.19%	10.28%	61.19%	20.08%	59.42%	8.30%	62.26%	11.49%	48,162
DCN	61.03%	8.58%	62.58%	17.40%	59.92%	7.82%	62.04%	10.23%	151,252
TSception	61.57%	11.04%	63.24%	16.60%	59.14%	7.60%	62.33%	9.03%	12,563

p-value between the method and TSception: * indicating ($p < 0.05$), ** indicating ($p < 0.01$), *** indicating ($p < 0.001$).

SCN: ShallowConvNet.

DCN: DeepConvNet.

coefficients achieved higher mean accuracy on validation set. The sampling rate of MAHNOB-HCI is 256Hz, which gives the temporal kernel lengths of 64, 32, and 16 as well. The number of temporal and spatial kernels in dynamic temporal, asymmetric spatial, and high-level fusion layers is equal to 15. The number of hidden nodes in the first fully connected layer is chosen as 32. For model training, the maximum training epoch is 500. The batch size on the DEAP dataset is set as 64 which will be reduced to 32 on the MAHNOB-HCI dataset because the trials in MAHNOB-HCI are half of the ones in DEAP. All the other hyper-parameters (including the structure hyper-parameters as well as the training hyper-parameters), except batch size, are the same for DEAP and MAHNOB-HCI to test the generalization ability of TSception. The hyper-parameters are the same for all the subjects. Adam optimizer is utilized to optimize the training process with the initial learning rate being $1e-3$. Cross-entropy loss is selected as the loss function to guide the training process. For more details, please refer to the open-access GitHub repository for TSception.

4 RESULTS AND ANALYSIS

In this section, we first report and statistically compare the results in terms of accuracy and F1 score for ours against the state-of-the-art methods. The ablation studies are then presented to reveal the contribution of each component in TSception. Finally, saliency maps are presented to visualize how the brain areas contribute to the arousal and valence dimensions.

4.1 Statistical Analysis

The experiment results: include I) the per-subject accuracy and F1 score on DEAP dataset (see Figs. 3 and 4), II) the overall accuracy and F1 score on DEAP dataset (see Table 3) MAHNOB-HCI data set (see Table 4), and III) the comparison against the results from existing literatures (see Table 5). To conduct statistical analysis, a two-tailed Wilcoxon Signed-Rank Test is utilized. Compared to accuracy, F1 score is a more reliable metric to quantify the performance of classification methods when a dataset has imbalanced classes. Based on the results we have the following observation and analysis.

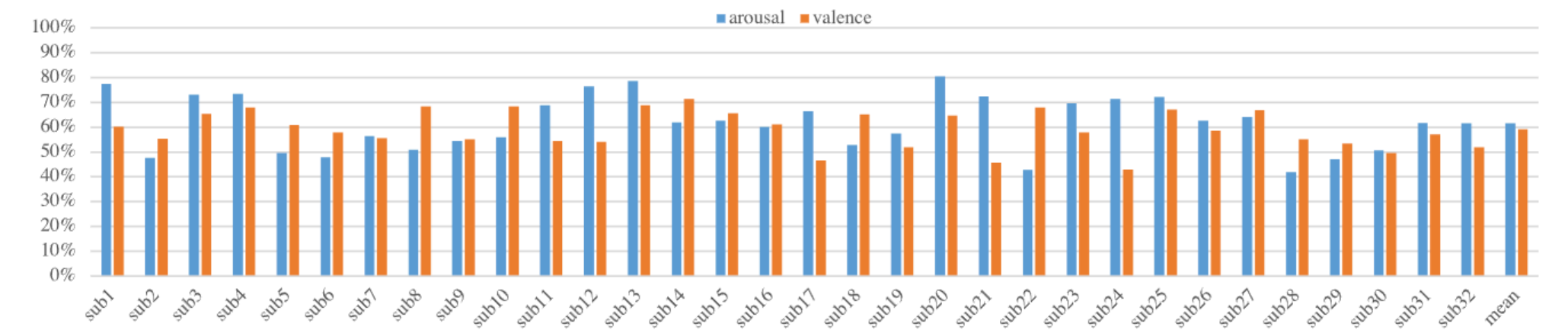


图 3. TSception 在 DEAP 上对每个受试者的唤醒度和效价平均准确率。

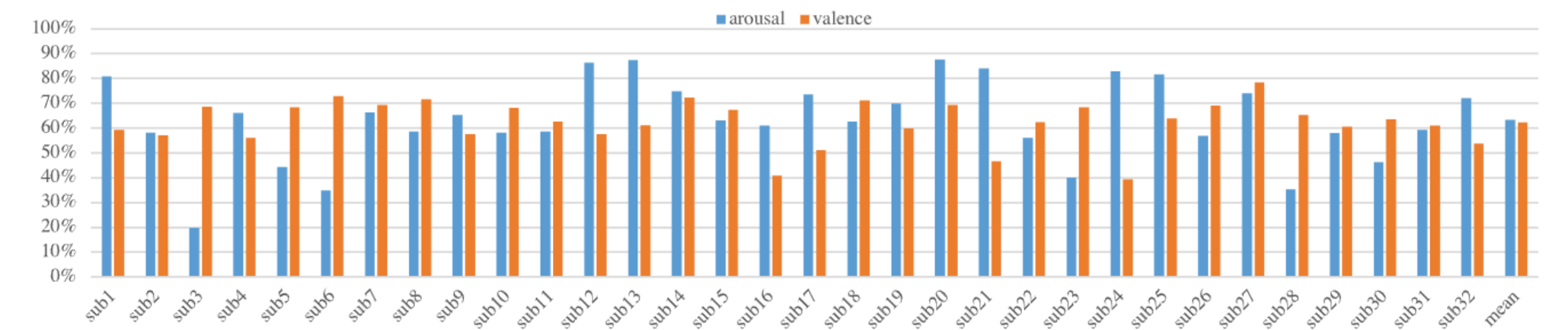


图 4. TSception 在 DEAP 上对每个受试者的唤醒度和效价平均 F1 分数。

表 3. SVM、KNN、EEGNet、ShallowConvNet、DeepConvNet 和 TSception 在 DEAP 上进行的试验级 10 折交叉验证分类结果。

唤醒度					效价				
方法	ACC	标准差	F1	标准差	ACC	标准差	F1	标准差	参数
SVM	60.37%	12.25%	57.33%	26.61%	55.19%	** 6.97%	57.87%**	11.36%	NA
KNN	59.48%	12.34%	57.49%	24.96%	53.03%	** 9.14%	55.12%**	16.27%	NA
EEGNet	58.29%	8.60%	60.60%	15.20%					
SCN	61.19%	10.28%	61.19%	20.08%	54.56% **	8.14%	57.61%**	10.42%	2,162
DCN	61.03%	8.58%	62.58%	17.40%	59.42%	8.30%	62.26%	11.49%	48,162
TSception	61.57%	11.04%	63.24%	16.60%	59.92%	7.82%	62.04%	10.23%	151,252
TSception	61.57%	11.04%	63.24%	16.60%	59.14%	7.60%	62.33%	9.03%	12,563

方法与 TSception 之间的 p 值：*表示 $\delta p < 0:05P$ ，**表示 $\delta p < 0:01P$ ，***表示 $\delta p < 0:001P$ 。
SCN: ShallowConvNet。
DCN: DeepConvNet。

系数在验证集上实现了更高的平均准确率。MAHNOB-HCI 的采样率为 256Hz，这也使得时间卷积核长度为 64、32 和 16。动态时间层、非对称空间层和高级融合层中的时间卷积核和空间卷积核数量均为 15。第一层全连接层的隐藏节点数选择为 32。模型训练中，最大训练轮数为 500。在 DEAP 数据集上，批大小设置为 64，在 MAHNOB-HCI 数据集上会减少到 32，因为 MAHNOB-HCI 中的试验数量是 DEAP 的一半。除了批大小之外，DEAP 和 MAHNOB-HCI 的所有其他超参数（包括结构超参数和训练超参数）都相同，以测试 TSception 的泛化能力。所有受试者的超参数都相同。使用 Adam 优化器优化训练过程，初始学习率为 1e-3。选择交叉熵损失函数来指导训练过程。更多细节请参考 TSception 的开源 GitHub 仓库。

4 结果与分析

在本节中，我们首先报告并统计比较了我们的方法与当前最先进方法在准确率和 F1 分数方面的结果。然后，我们展示了消融研究，以揭示 TSception 中每个组件的贡献。最后，我们展示了显著性图，以可视化大脑区域如何对唤醒度和效价维度做出贡献。

4.1 统计分析

实验结果包括：I) DEAP 数据集上每个受试者的准确率和 F1 分数（见图 3 和图 4），II) DEAP 数据集和 MAHNOB-HCI 数据集上的总体准确率和 F1 分数（见表 3 和表 4），以及 III) 与现有文献结果的比较（见表 5）。为了进行统计分析，我们使用了双尾 Wilcoxon 符号秩检验。与准确率相比，当数据集类别不平衡时，F1 分数是衡量分类方法性能的更可靠的指标。根据我们的结果，我们得出以下观察和分析。

TABLE 4
Trial-Wise 10-Fold Cross-Validation Classification Results of SVM, KNN, EEGNet, ShallowConvNet, DeepConvNet, TSception on MAHNOB-HCI

Method	Arousal				Valence				Parameters
	ACC	std	F1	std	ACC	std	F1	std	
SVM	58.25% *	14.09%	33.40%	21.87%	58.44% *	9.39%	40.27%	12.73%	NA
KNN	60.95%	17.11%	28.65%	24.94%	60.32%	12.28%	28.62%***	20.26%	NA
EEGNet	59.98%	16.16%	30.47%	23.68%	56.43%***	11.12%	33.98%**	15.27%	2,674
SCN	59.85%	16.02%	30.60%	22.20%	59.57%	11.25%	36.41%**	15.27%	50,882
DCN	57.29%	15.69%	32.37%	23.69%	60.29%	12.38%	36.09%*	17.64%	153,652
TSception	60.61%	14.88%	33.06%	23.35%	61.27%	10.05%	40.66%	16.52%	12,563

p-value between the method and TSception: * indicating ($p < 0.05$), ** indicating ($p < 0.01$), *** indicating ($p < 0.001$).

SCN: ShallowConvNet.

DCN: DeepConvNet.

On DEAP, deep learning methods generally outperform non-deep learning methods, whereas, on MAHNOB-HCI, the SVM outperforms EEGNet, ShallowConvNet, and DeepConvNet, and is comparable to our method. On DEAP, ours outperforms SVM and KNN on all the experiments. Ours has a 1.2% higher accuracy and a 5.91% higher F1 score than SVM for the arousal dimension. For valence, ours outperform SVM with the improvement in accuracy and F1 score being 3.95% ($p < 0.05$) and 4.46% ($p < 0.05$) respectively. Compared with KNN, ours achieves 2.09%/5.91% higher accuracy/F1 score for arousal and 3.95% ($p < 0.05$)/7.21% ($p < 0.05$) higher accuracy/F1 score for valence. On MAHNOB-HCI, ours achieves the best accuracy and F1 score on valence, while the best accuracy on arousal is achieved by KNN classifier and the best F1 score is the one using SVM. But TSception still achieves a 4.41% higher F1 score than KNN ($p = 0.07186$) and a 2.36% higher accuracy than SVM ($p < 0.05$) for arousal dimension. Although the other three deep learning methods have higher classification results the SVM and KNN on DEAP, they have lower accuracy or F1 score than those non-deep learning methods. SVM defeats EEGNet, ShallowConvNet, and DeepConvNet on F1 scores for both arousal and valence dimensions. And KNN achieves better accuracies than those three deep learning methods for both arousal and valence. It further suggests that except for ours, most of the deep learning methods demonstrate less cross-dataset generality comparing to SVM and KNN.

Among the four deep learning methods, ours achieves the highest accuracy and F1 score in most of the experiments on DEAP. In particular, our TSception (12,563) has only a quarter and one-tenth of the parameters compared to ShallowConvNet (48,162) and DeepConvNet (151,252), respectively. TSception achieves the highest accuracy and F1 score on arousal as well as the highest F1 score on valence, with the accuracy being 61.57% for arousal, 59.14% for valence, and the F1 score being 63.24% for arousal, 62.33% for valence respectively. DeepConvNet achieves second place compared with the other methods (accuracy: 61.03% for arousal, 59.92% for valence, F1: 62.58% for arousal, 62.04% for valence). ShallowConvNet gets the third-highest results among all the compared methods, achieving 61.19%/59.42% for arousal/valence in terms of accuracy, and 61.19%/62.26% for arousal/valence in terms of F1 score. The accuracy of TSception for arousal is 3.28% higher than EEGNet ($p = 0.05118$), the one for valence has a

4.58% improvement over EEGNet ($p < 0.01$). For the F1 score, TSception has a 2.64% higher F1 score than EEGNet for arousal ($p = 0.05614$), and a 5.72% higher F1 score for valence ($p < 0.01$).

On MAHNOB-HCI, ours achieves the highest accuracy and F1 score among four deep learning methods. For accuracy metrics, TSception is 3.32% ($p = 0.0536$) and 0.98% ($p = 0.35238$) higher than DeepConvNet for arousal and valence respectively. Compared with ShallowConvNet, TSception has higher accuracies for arousal (0.76%) and valence (1.7%) with the p -value being 0.48392 and 0.12356. TSception is 0.63% ($p = 0.87288$) and 4.84% ($p < 0.001$) higher than EEGNet for arousal and valence in terms of accuracy. For F1 scores, TSception achieves much higher results than the other three deep learning methods. Especially for valence dimension, TSception achieves 4.57% ($p < 0.05$), 4.25% ($p < 0.01$), and 6.68% ($p < 0.01$) higher F1 score than DeepConvNet, ShallowConvNet, and EEGNet.

Interestingly, we also notice that the difficulty to predict the two emotional dimensions are not consistent for the two datasets. Considering the trade-off of accuracy and F1 score, we find that the valence is harder to predict for DEAP while the arousal is harder to predict for MAHNOB-HCI.

Our method outperforms the results reported in the existing literatures [33], [38], [39] as well. According to Table 5, ours achieves the best accuracies for both arousal and valence dimensions. TSception has 3.15% and 1.18% improvements over FBTSC [39] on accuracy for arousal and valence. Compared with UL [38], our method beats it by

TABLE 5
Compare with the Results Reported in the Existing Literatures Using Leave-One-Trial-Out Cross-Validation on DEAP

Method	Arousal		Valence	
	ACC	F1	ACC	F1
SVM[33]	62.00%	58.30%	57.60%	56.30%
UL[38]	62.34%	60.44%	56.25%	61.25%
CSP[39]	58.26%	-	57.59%	-
FBCSP[39]	59.13%	-	59.19%	-
FgMDM[39]	60.04%	-	58.87%	-
TSC[39]	60.04%	-	59.47%	-
FBFgMDM[39]	60.30%	-	61.01%	-
FBTSC[39]	60.60%	-	61.09%	-
TSception(ours)	63.75%	63.35%	62.27%	65.37%

表 4 MAHNOB-HCI 上 SVM、KNN、EEGNet、ShallowConvNet、DeepConvNet、TSception 的试验级 10 折交叉验证分类结果

唤醒度					效价				
方法	ACC	std	F1	std	ACC	std	F1	std	参数
SVM	58.25%	*	14.09%	33.40%	21.87%	58.44%	*	9.39%	40.27%
EEGNet	12.73%	NA	KNN	60.95%	17.11%	28.65%	24.94%	60.32%	12.28%
DCN	57.29%	15.69%	32.37%	23.69%	60.29%	12.38%	36.09%*	17.64%	153,652
TSception	60.61%	14.88%	33.06%	23.35%	61.27%	10.05%	40.66%	16.52%	12,563

方法与 TSception 之间的 p 值：*表示 (p < 0:05)，**表示 (p < 0:01)，***表示 (p < 0:001)。
SCN: ShallowConvNet。
DCN: DeepConvNet。

在 DEAP 数据集上，深度学习方法通常优于非深度学习方法，而在 MAHNOB-HCI 数据集上，SVM 的表现优于 EEGNet、ShallowConvNet 和 DeepConvNet，并且与我们方法相当。在 DEAP 数据集上，我们的方法在所有实验中都优于 SVM 和 KNN。对于唤醒维度，我们的方法比 SVM 的准确率高出 1.2%，F1 分数高出 5.91%。对于效价维度，我们的方法比 SVM 的准确率提高了 3.95% (p < 0:05)，F1 分数提高了 4.46% (p < 0:05)。与 KNN 相比，我们的方法在唤醒维度上实现了 2.09%/5.91%更高的准确率/F1 分数，在效价维度上实现了 3.95% (p < 0:05) /7.21% (p < 0:05) 更高的准确率/F1 分数。在 MAHNOB-HCI 数据集上，我们的方法在效价维度上获得了最高的准确率和 F1 分数，而唤醒维度上最高的准确率由 KNN 分类器获得，最高的 F1 分数则使用 SVM 实现。但 TSception 在唤醒维度上仍然比 KNN 高出 4.41%的 F1 分数 (p ¼ 0:07186)，比 SVM 高出 2.36%的准确率 (p < 0:05)。尽管其他三种深度学习方法在 DEAP 数据集上的分类结果更高，但它们的准确率或 F1 分数仍低于这些非深度学习方法。SVM 在唤醒度和效价两个维度上的 F1 分数均优于 EEGNet、ShallowConvNet 和 DeepConvNet。而 KNN 在唤醒度和效价两个维度上的准确率均高于这三种深度学习方法。这进一步表明，除了我们的方法外，大多数深度学习方法在跨数据集泛化能力上均不如 SVM 和 KNN。在四种深度学习方法中，我们的方法在 DEAP 数据集上的大多数实验中实现了最高的准确率和 F1 分数。特别是，我们的 TSception (12,563) 与 ShallowConvNet (48,162) 和 DeepConvNet (151,252) 相比，参数数量分别只有四分之一和十分之一。TSception 在唤醒度上实现了最高的准确率和 F1 分数，在效价上实现了最高的 F1 分数，唤醒度的准确率为 61.57%，效价的准确率为 59.14%，唤醒度的 F1 分数为 63.24%，效价的 F1 分数为 62.33%。与其他方法相比，DeepConvNet 位居第二（唤醒度的准确率为 61.03%，效价的准确率为 59.92%，唤醒度的 F1 分数为 62.58%，效价的 F1 分数为 62.04%）。ShallowConvNet 在所有比较方法中取得了第三高的结果，在唤醒度/效价上的准确率分别为 61.19%/59.42%，在 F1 分数上的唤醒度/效价分别为 61.19%/62.26%。TSception 在唤醒度上的准确率比 EEGNet 高 3.28% (p=0.05118)。

效价方面的模型比 EEGNet 提高了 4.58% (p < 0:01)。在 F1 分数方面，TSception 在唤醒度上的 F1 分数比 EEGNet 高 2.64% (p ¼ 0:05614)，在效价上的 F1 分数比 EEGNet 高 5.72% (p < 0:01)。

在 MAHNOB-HCI 上，我们的方法在四种深度学习方法中实现了最高的准确率和 F1 分数。在准确率指标上，TSception 在唤醒度和效价方面分别比 DeepConvNet 高 3.32% (p=0.0536) 和 0.98% (p=0.35238)。与 ShallowConvNet 相比，TSception 在唤醒度 (0.76%) 和效价 (1.7%) 上的准确率更高，p 值分别为 0.48392 和 0.12356。在准确率方面，TSception 在唤醒度和效价上分别比 EEGNet 高 0.63% (p=0.87288) 和 4.84% (p<0.001)。在 F1 分数上，TSception 的结果明显高于其他三种深度学习方法。特别是在效价维度上，TSception 的 F1 分数分别比 DeepConvNet、ShallowConvNet 和 EEGNet 高 4.57% (p<0.05)、4.25% (p<0.01) 和 6.68% (p<0.01)。

有趣的是，我们还注意到预测这两个情绪维度在两个数据集上的难度并不一致。考虑到准确率和 F1 分数之间的权衡，我们发现效价在 DEAP 数据集上更难预测，而唤醒度在 MAHNOB-HCI 数据集上更难预测。

我们的方法在现有文献[33]、[38]、[39]中报告的结果上也表现更优。根据表 5，我们在唤醒度和效价维度上均取得了最高的准确率。TSception 在唤醒度和效价准确率上分别比 FBTSC[39]提高了 3.15%和 1.18%。与 UL[38]相比，我们的方法在

表 5 使用 Leave-One-Trial-Out 交叉验证在 DEAP 上与现有文献报道的结果进行比较

方法	唤醒		效价	
	ACC	F1	ACC	F1
SVM[33]	62.00%	58.30%	57.60%	56.30%
UL[38]	62.34%	60.44%	56.25%	61.25%
CSP[39]	58.26%	-	57.59%	-
FBCSP[39]	59.13%	-	59.19%	-
FgMDM[39]	60.04%	-	58.87%	-
TSC[39]	60.04%	-	59.47%	-
FBFgMDM[39]	60.30%	-	61.01%	-
FBTSC[39]	60.60%	-	61.09%	-
TSception(我们)	63.75%	63.35%	62.27%	65.37%

TABLE 6
Ablation Study Results of Removing Functional Layers in TSception Using DEAP

Method	Arousal		Valence	
	ACC	F1	ACC	F1
w/o T	60.45%	61.29%	58.62%	61.47%
w/o S	60.07%	61.41%	56.90%	60.74%
w/o F	60.03%	60.85%	58.21%	61.14%
TSception	61.57%	63.25%	59.14%	62.33%

T: Dynamic temporal layer; S: Asymmetric spatial layer; F: High-level fusion layer.

w/o: Without the component.

1.41% for arousal and 6.02% for valence in terms of accuracy. The accuracies of ours for arousal and valence are 1.75% and 4.67% higher than the ones of SVM reported in [33]. For F1 scores, TSception has 5.05% and 2.91% higher than SVM [33] and UL [38] for arousal and 9.07% and 4.12% higher for valence, indicating the effectiveness of the proposed method.

According to the extensive comparison against a variety of methods, the proposed method manifests promising performance on the arousal-valence prediction task, with a decent extent of generality.

4.2 Ablation Study

The proposed method TSception has a dynamic temporal layer, asymmetric spatial layer, and high-level fusion layer three functional parts. The combination of those three parts leads to the success of classification tasks. Ablation studies are conducted to further understand which part contributes more to the improvement of classification results. The classification results after removing each of the dynamic temporal layer, asymmetric spatial layer, and high-level fusion layer from the TSception are reported. DEAP dataset is used for the ablation study since the overall performance is higher than MAHNOB-HCI. The results of the ablation study are shown in Table 6.

All of the accuracies and F1 scores drop after removing any of the three types of layers, indicating all components contribute to the improvement of classification results. Overall, the most significant drops of accuracy for three dimensions are observed when the asymmetric spatial layer is removed from TSception with the decrements being 1.5%/1.84% on accuracies for arousal/valence and 1.84%/1.59% on F1 scores for arousal/valence. This demonstrated that the asymmetric spatial layer contributes more than the other two layers, especially for the valence dimension, the drop is the largest in the ablation study. The high-level fusion layer contributes more to arousal because the accuracy drops by 1.54%, and the F1 score drops by 2.40% for arousal while the drops of accuracy and F1 score for valence are smaller (0.93% on accuracy and 1.19% on F1 score) after removing the high-level fusion layer. The dynamic temporal layer contributes less than the others, with the drops of accuracy/F1 score being 1.12%/1.96% for arousal and 0.52%/0.86% for valence.

The kernel-level ablation studies are further conducted to analyze the effects of two types of spatial kernels in the spatial asymmetric layer because it has more contribution than

TABLE 7
Ablation Study Results of Removing Spatial Convolutional Kernels in TSception Using DEAP

Method	Arousal		Valence	
	ACC	F1	ACC	F1
w/o H	55.86%	50.38%	51.15%	40.13%
w/o G	57.21%	58.29%	54.22%	57.75%
TSception	61.57%	63.25%	59.14%	62.33%

H: Hemisphere kernels; G: Global kernels.

w/o: Without the component.

other layers. The weights and biases are set to zeros as [18] did to study the kernel-level effects. The results are shown in Table 7.

Hemisphere kernels learn more discriminative representations than global kernels in TSception, according to the results in Table 7. The drops of classification results after removing the hemisphere kernels are all larger than the ones after removing the global kernels. After removing either type of the hemisphere and global kernels will downgrade the performance of TSception for both the arousal and valence dimensions. This indicates both types of spatial convolutions help to improve the performance of TSception.

4.3 Interpretability

In this part, the saliency map [45] is utilized to visualize which parts of the data are more informative and contribute to classification performance. The saliency map is one of the most commonly used tools to intuitively show which regions of the input have the classification-related information. To better visualize the saliency map, the original saliency map is averaged along the time dimension to get the topological map of the EEG channels. The normalized saliency maps of different samples of each subject are averaged to get the mean saliency map of the subject for general visualization. The averaged saliency maps in the DEAP dataset are shown in Fig. 5. The mean saliency maps of individuals for arousal are also shown in Fig. 6 to illustrate the differences across subjects.

The pictures in Fig. 5 are the saliency maps under different calculation settings. The upper three saliency maps, Figs. 5a, 5b, and 5c, are the averaged saliency maps for arousal dimension while the lower three, Figs. 5d, 5e, and 5f, are for valence. The first column, Figs. 5a and 5d, are the mean saliency map of all the subjects. The second column, Figs. 5b and 5e, are the one of subjects who are top 10% for F1 scores. The last column, Figs. 5c and 5f, are the average saliency map of the subjects whose F1 scores are in bottom 10% for arousal. The mean saliency map is normalized between -1 and 1 for better visualization. We choose F1 as the selecting criterion for visualization because it can reflect how precise the predictions are when the classes are imbalanced.

For arousal, the frontal, temporal, and right side of the parietal and occipital areas of the brain are more informative according to Figs. 5a and 5b. The averaged saliency map of all the subjects, Fig. 5a, shows the value of Fp2, F3, FC2, FC5, T7, T8, C4, P8, and O2 channels are higher than others. Comparing the saliency maps of the top (Fig. 5b) and bottom 10% (Fig. 5c) F1 score subjects, we can see the frontal (Fp1, AF3, F3 and F4), temporal (T8) and parietal (P7) areas

表 6 移除功能模块的消融研究结果

TSception 中使用的 DEAP 层				
方法	唤醒		效价	
	ACC	F1	ACC	F1
无 T	60.45%	61.29%	58.62%	61.47%
无 S	60.07%	61.41%	56.90%	60.74%
不包含 F	60.03%	60.85%	58.21%	61.14%
TSception	61.57%	63.25%	59.14%	62.33%

T: 动态时间层; S: 非对称空间层; F: 高级融合层。

w/o: 不包含该组件。

在唤醒度方面准确率为 1.41%，在效价方面准确率为 6.02%。我们的唤醒度和效价准确率分别比[33]中报告的 SVM 高 1.75%和 4.67%。在 F1 分数方面，TSception 在唤醒度上比 SVM[33]和 UL[38]分别高 5.05%和 2.91%，在效价上分别高 9.07%和 4.12%，这表明了所提出方法的有效性。

根据与各种方法的广泛比较，所提出的方法在唤醒-效价预测任务上表现出良好的性能，具有相当程度的泛化能力。

4.2 消融实验

所提出的方法 TSception 包含动态时间层、非对称空间层和高级融合层三个功能部分。这三个部分的结合促成了分类任务的成功。通过消融实验进一步了解哪个部分对分类结果的改进贡献更大。从 TSception 中移除动态时间层、非对称空间层和高级融合层后，分别报告了分类结果。由于整体性能高于 MAHNOB-HCI，因此使用 DEAP 数据集进行消融实验。消融实验的结果如表 6 所示。

移除任何一种类型的层后，所有准确率和 F1 分数都会下降，这表明所有组件都对分类结果的改进有贡献。总体而言，当从 TSception 中移除非对称空间层时，三个维度上的准确率下降最为显著，具体表现为唤醒度/效价准确率分别下降了 1.5%/1.84%，F1 分数分别下降了 1.84%/1.59%。这表明非对称空间层比其他两层贡献更大，尤其是在效价维度上，消融研究中的下降幅度最大。高层融合层对唤醒度贡献更大，因为移除该层后，唤醒度准确率下降了 1.54%，F1 分数下降了 2.40%，而效价准确率和 F1 分数的下降幅度较小（准确率下降 0.93%，F1 分数下降 1.19%）。动态时间层贡献较小，移除该层后，唤醒度准确率/F1 分数下降了 1.12%/1.96%，效价准确率/F1 分数下降了 0.52%/0.86%。

在内核层进行消融研究，进一步分析了空间非对称层中两种空间核的影响，因为它们比其他因素有更大的贡献

表 7 移除空间卷积的消融研究结果

TSception 中使用 DEAP 的核函数				
方法	唤醒		效价	
	ACC	F1	ACC	F1
无 H	55.86%	50.38%	51.15%	40.13%
无 G	57.21%	58.29%	54.22%	57.75%
TSception	61.57%	63.25%	59.14%	62.33%

H: 半球核; G: 全局核。
w/o: 无该成分。

其他层。权重和偏差被设置为 0，如[18]所做，以研究内核级效应。结果如表 7 所示。

半球核在 TSception 中比全局核学习到更具判别性的表示，如表 7 中的结果所示。移除半球核后分类结果的下降幅度都大于移除全局核后的下降幅度。移除半球核或全局核中的任何一种都会降低 TSception 在唤醒度和效价维度上的性能。这表明这两种空间卷积都有助于提高 TSception 的性能。

4.3 可解释性

在这一部分，使用显著性图[45]来可视化数据中哪些部分更具信息量，并有助于分类性能。显著性图是最常用的工具之一，可以直观地显示输入中哪些区域包含分类相关信息。为了更好地可视化显著性图，原始显著性图沿时间维度进行平均，得到 EEG 通道的拓扑图。将每个受试者不同样本的归一化显著性图进行平均，得到受试者的平均显著性图，用于一般可视化。DEAP 数据集中的平均显著性图显示在图 5 中。为了说明受试者之间的差异，个体的平均显著性图（针对唤醒度）也显示在图 6 中。

图 5 中的图片是在不同计算设置下的显著性图。上方的三个显著性图，即图 5a、5b 和 5c，是唤醒维度的平均显著性图，而下方三个，即图 5d、5e 和 5f，是效价维度的。第一列，图 5a 和 5d，是所有受试者的平均显著性图。第二列，图 5b 和 5e，是 F1 分数排名前 10%受试者的平均显著性图。最后一列，图 5c 和 5f，是唤醒维度 F1 分数排名后 10%受试者的平均显著性图。平均显著性图在-1 到 1 之间进行了归一化，以便更好地可视化。我们选择 F1 作为可视化选择标准，因为它可以反映在类别不平衡时预测的精确度。

对于唤醒度，根据图 5a 和图 5b，大脑的前额叶、颞叶以及顶叶和枕叶的右侧区域更具信息量。所有受试者的平均显著性图（图 5a）显示，Fp2、F3、FC2、FC5、T7、T8、C4、P8 和 O2 通道的值高于其他通道。比较得分最高的前 10%（图 5b）和最低的 10%（图 5c）受试者的显著性图，我们可以看到前额叶（Fp1、AF3、F3 和 F4）、颞叶（T8）和顶叶（P7）区域

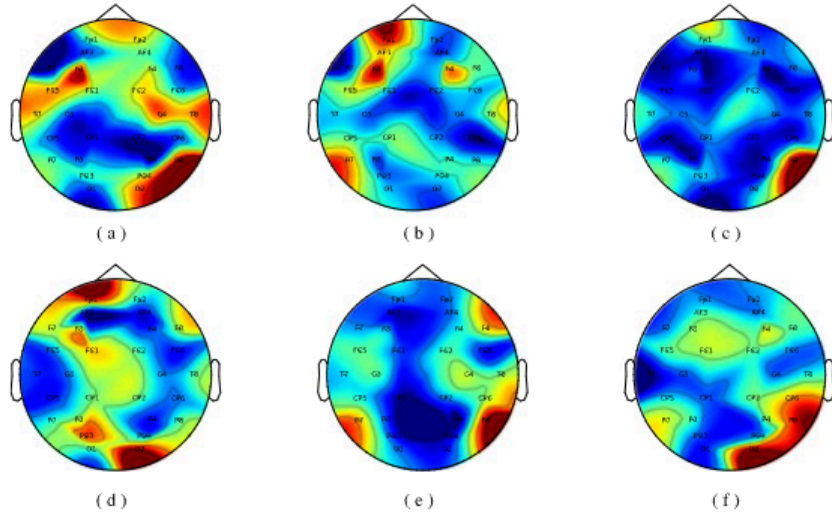


Fig. 5. Averaged saliency maps in DEAP dataset. The upper three saliency maps (a)-(c) are the averaged saliency maps for arousal dimension while the lower three (d)-(f) are for valence. The first column (a) and (d) are the mean saliency map of all the subjects. The second column (b) and (e) are the one of subjects who are top 10% for F1 scores, The last column (c) and (f) are the average saliency map of the subjects whose F1 scores are in bottom 10% for arousal. The mean saliency map is normalized between -1 and 1 for better visualization. F1 is chosen as the criterion because it can reflect how precise the predictions are by taking the imbalanced classes issue into consideration. The most informative region identified by the neural network is the frontal, temporal, parietal, and regions for high F1 score subjects.

provide more information in Fig. 5b, while the network mainly learns from parietal (P8) in Fig. 5c. This indicates frontal, temporal, and parietal areas of the brain provide more emotion-related information. This is consistent with previous literatures [46], [47], [48]. Emotion arousal is mostly reflected in the frontal lobe, temporal lobe, and parietal lobe [46]. Pre-frontal and temporal asymmetry have close relations to arousal recognition [47].

For valence, the frontal, temporal, and right side of the parietal and occipital areas of the brain are more informative according to Figs. 5d and 5e. The same thing happens that the

occipital (O1 and O2) activities provide less classification-related information than frontal (F8), temporal (T8), and parietal (P7 and P8) activities for the high F1 score subjects (Fig. 5e). According to previous studies, the asymmetry patterns in pre-frontal, parietal, and temporal regions are observed for valence recognition [47].

In general, the most informative region identified by the neural network is the frontal, temporal, parietal, and occipital regions while the occipital activities are less informative for the subjects with high F1 scores. This is consistent with previous works [44], [46], [47], [49], [50], which indicates the network learns from the proper region. And for both arousal and valence, the occipital activities provide certain information. This may be because the stimuli used in DEAP are music videos.

5 DISCUSSION AND CONCLUSION

Accurate emotion detection can benefit many healthcare applications including Cognitive Behavioural Therapy (CBT), Emotion Regulation Therapy (ERT)/Emotion-Focused Therapy (EFT) for emotion-related mental disorder treatment. Most of the previous works highly rely on the human extracted features, which requires heavy domain knowledge. Deep learning, especially the family of convolutional neural networks, has the auto feature-extracting ability. In this paper, we propose TSception, a multi-scale convolutional neural network, for EEG emotion recognition tasks. The parallel multi-scale temporal kernels whose lengths are related to the sampling rate of EEG are proposed in the temporal convolutional layer of TSception to enrich the learned temporal/frequency representations. To capture the emotional asymmetry patterns, we propose hemisphere kernels besides the global

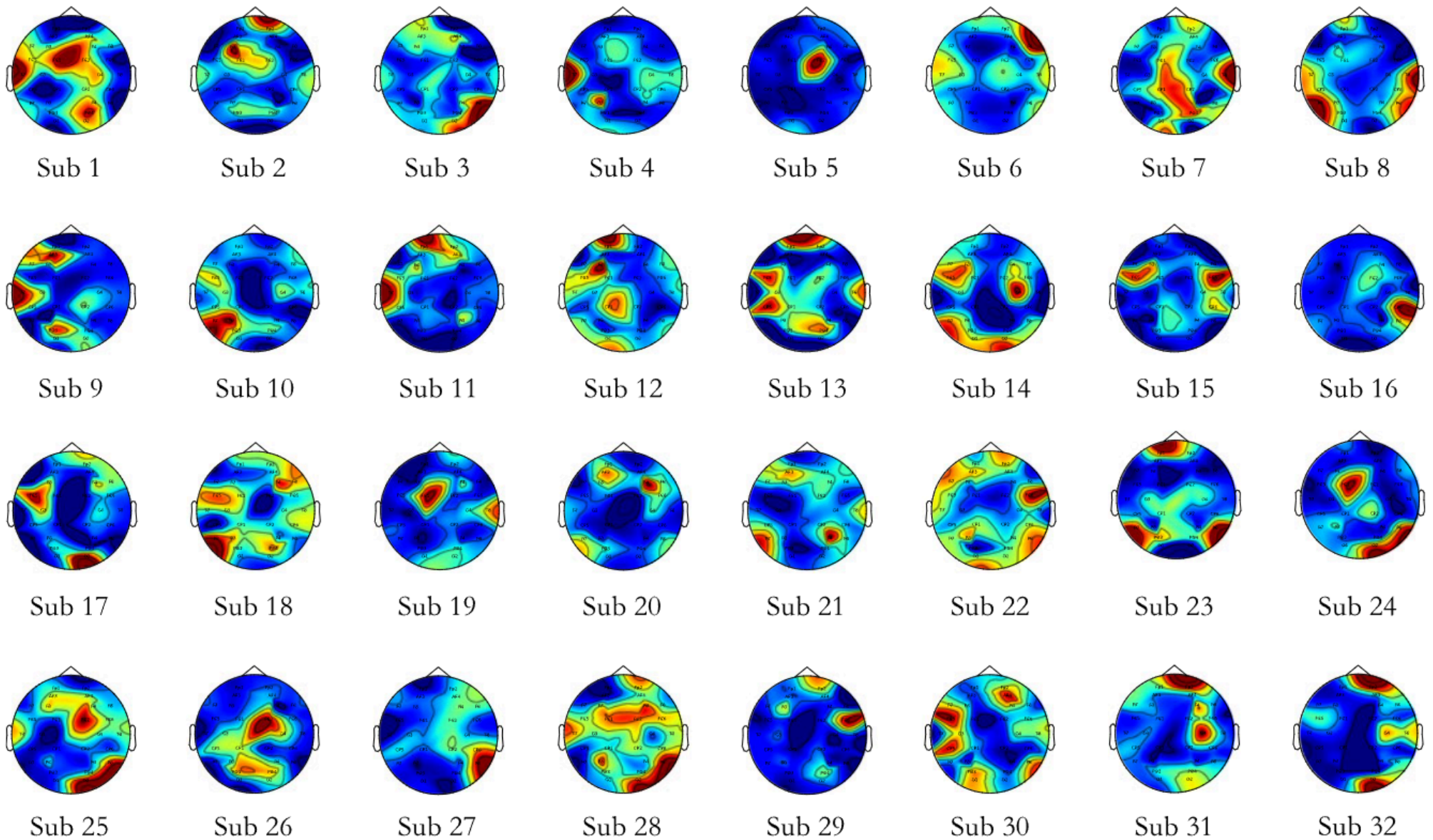


Fig. 6. Saliency maps of all 32 subjects for arousal in DEAP dataset. The saliency map is averaged along the time dimension to plot the topological map.

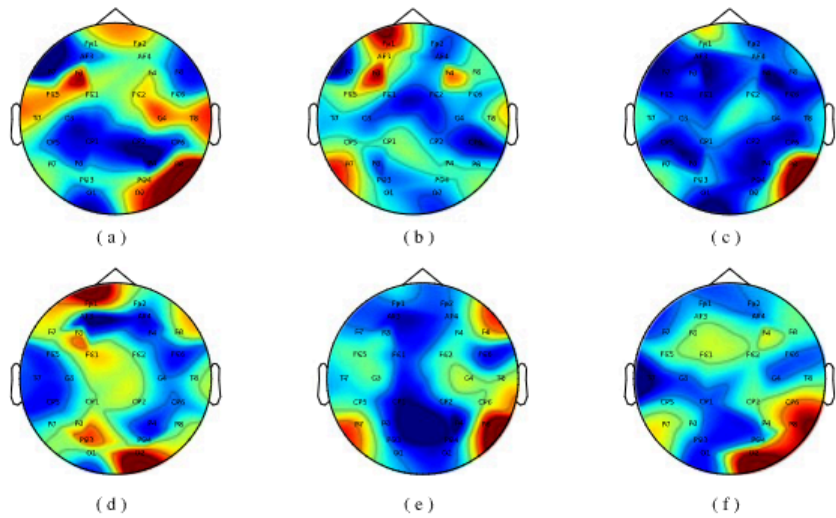


图 5. DEAP 数据集的平均显著性图。上三个显著性图 (a) - (c) 是唤醒维度的平均显著性图，下三个 (d) - (f) 是效价维度的。第一列 (a) 和 (d) 是所有受试者的平均显著性图。第二列 (b) 和 (e) 是 F1 分数排名前 10% 的受试者的平均显著性图，最后一列 (c) 和 (f) 是唤醒维度 F1 分数排名后 10% 的受试者的平均显著性图。平均显著性图在 -1 到 1 之间进行了归一化，以便更好地可视化。选择 F1 作为标准，因为它考虑了类别不平衡问题，能够反映预测的精确度。神经网络识别的最信息量大的区域是额叶、颞叶、顶叶以及高 F1 分数受试者的区域。

P8 在图 5b 中提供更多信息，而网络主要从顶叶 (P8) 学习，如图 5c 所示。这表明大脑的前额叶、颞叶和顶叶区域提供了更多与情绪相关的信息。这与先前文献[46]、[47]、[48]一致。情绪唤醒主要体现在前额叶、颞叶和顶叶[46]。前额叶和颞叶的不对称性与唤醒识别密切相关[47]。

对于效价，根据图 5d 和 5e，大脑的前额叶、颞叶以及顶叶和枕叶的右侧区域更具信息量

对于高 F1 分数的被试，同样存在这种情况：枕叶 (O1 和 O2) 的活动提供的分类相关信息少于额叶 (F8)、颞叶 (T8) 和顶叶 (P7 和 P8) 的活动 (图 5e)。根据先前的研究，在效价识别中观察到前额叶、顶叶和颞叶区域的不对称模式[47]。

总的来说，神经网络识别的最具信息量的区域是额叶、颞叶、顶叶和枕叶区域，而对于高 F1 分数的被试，枕叶活动信息量较少。这与先前的工作[44]、[46]、[47]、[49]、[50]一致，这表明网络从适当的区域学习。并且对于唤醒度和效价，枕叶活动都提供了一定的信息。这可能是因为 DEAP 中使用的刺激是音乐视频。

5 讨论和结论

准确的情感检测可以受益于许多医疗应用，包括认知行为疗法 (CBT)、情绪调节疗法 (ERT) /情绪聚焦疗法 (EFT) 等用于治疗与情感相关的心理疾病。以往的大多数工作高度依赖人工提取的特征，这需要大量的领域知识。深度学习，特别是卷积神经网络家族，具有自动特征提取的能力。在本文中，我们提出了 TSception，一种多尺度卷积神经网络，用于脑电图 (EEG) 情感识别任务。我们在 TSception 的时间卷积层中提出了并行多尺度时间卷积核，其长度与脑电图 (EEG) 的采样率相关，以丰富学习到的时间/频率表示。为了捕捉情感不对称模式，我们在全局卷积核之外提出了半球卷积核。

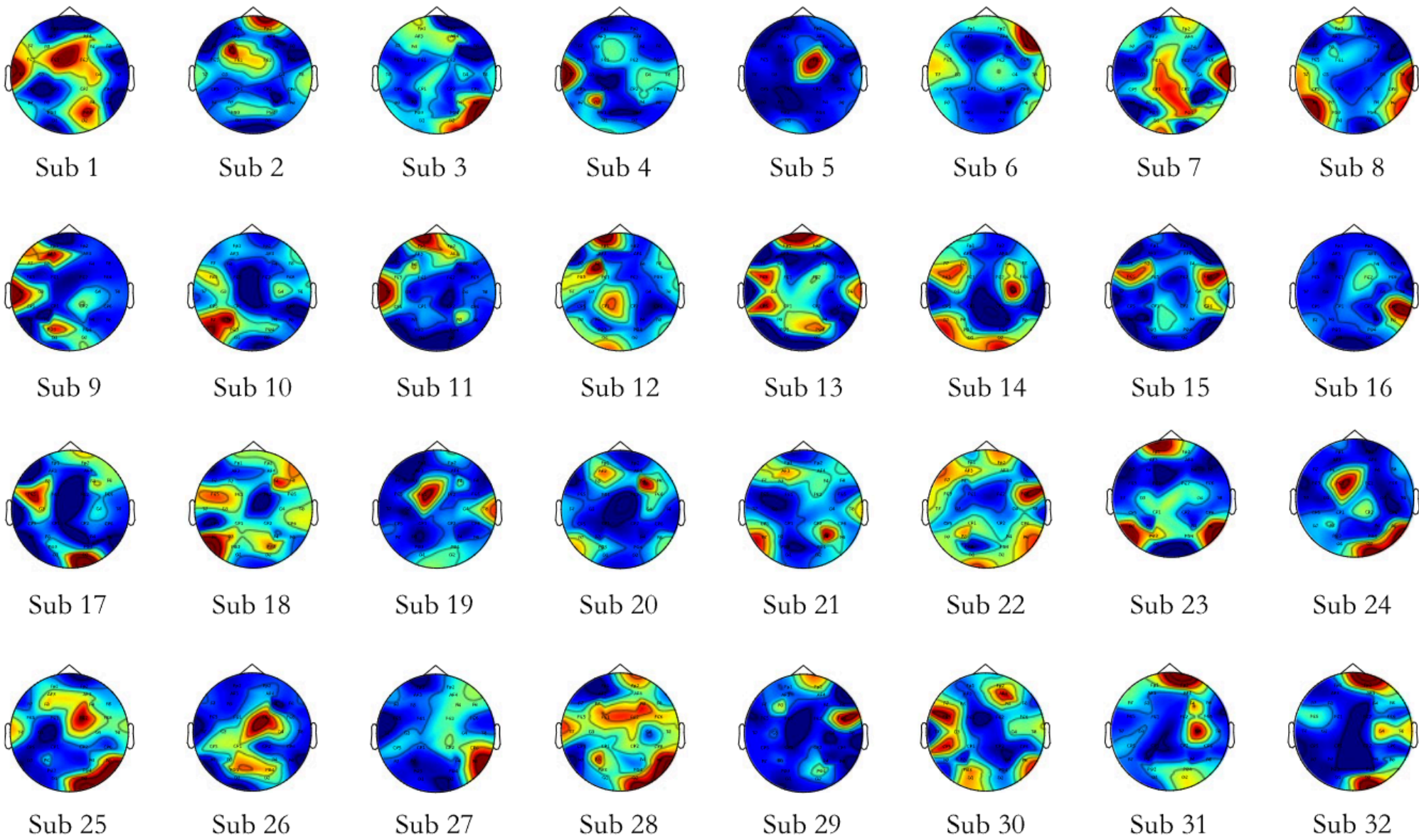


图 6. DEAP 数据集中所有 32 名受试者唤醒维度的显著性图。显著性图沿时间维度进行平均，以绘制拓扑图。

kernels in the asymmetric spatial layer. A high-level fusion layer is designed to further learn from the hemisphere/global representations of EEG and reduce the model size.

To get the generalized evaluation of our method, we adopt the trial-wise cross-validation of cropped trials on two benchmark datasets. As mentioned in Section 4.4, if one randomly shuffles the samples among different trials before dividing the data into training and testing data in cropped experiments, he can get very high classification results that will drop when the highly correlated adjacent segments in one trial are not seen by the model [38], [51]. Hence, the trial-wise 10-fold cross-validation is utilized to make sure the highly correlated adjacent segments of each trial don't appear in both training and testing data. To further compare our methods with the ones in the existing literatures that also use generalized evaluation settings, a leave-one-trial-out cross-validation is conducted with a voting mechanism on each trial's segment predictions. As for evaluating metrics, we also follow [38], adding F1 score besides accuracy to get a better evaluation on imbalanced datasets.

According to the results on two public datasets shown in Tables 3, 4, and 5, the proposed TSception achieves the highest classification results than those from the compared methods in most of the experiments. Particularly, TSception has 1/4 or 1/10 of the trainable parameters of its counterparts. Such efficiency and effectiveness may benefit the online usage of the neural network in real-world BCI applications.

Extensive ablation studies and interpretability experiments suggested that all modules in TSception have positive contributions to the improvement of classification results and our method learns from the emotion-related information. According to Table 6, we find the asymmetric spatial layer contributes most to the classification results. To make sure the neural network learns the emotion-related information instead of irrelevant features, saliency maps are acquired to visualize the most informative regions identified by the neural network itself. The mean saliency maps of all subjects in Figs. 5a and 5d show strong activation in the frontal, temporal, parietal, and occipital areas. However, the saliency maps of the subjects with high F1 scores in Figs. 5b and 5e only show strong activation in the frontal, temporal, and parietal areas, which is consistent with [1], [44], [46], [47], [49], [50], [52]. A right hemisphere lateralization pattern is also observed in the averaged saliency map of the top 10% subjects with high F1 scores (Fig. 5e) for valence, which indicates the right hemisphere is more informative for valence recognition. Neuroscience studies [53], [54] suggested that the right hemisphere has a special role in the emotional process in the brain. However, the right hemisphere lateralization is not present for high F1 subjects for arousal as shown in Fig. 5b. This could be because the information provided in the frontal area is enough for the neural network to make the decision. Moreover, we find that the occipital activities also contribute to the inference process of the neural network for all the subjects, as shown in Figs. 5a and 5d. A possible reason for high occipital activities is that music videos are used as stimuli in DEAP. However, the information provided by occipital activities is less useful for high F1 subjects (for both arousal and valence). This suggests occipital is not as informative as other brain regions, such as frontal and temporal regions, for emotion recognition.

To conclude, we propose a multi-scale convolutional neural network, named TSception, to capture temporal dynamics and spatial asymmetry for EEG emotion recognition. Using generalized cross-validation strategies, the proposed method and several baseline methods are evaluated on two publicly available benchmark datasets. The proposed method manifests promising performance on the arousal-valence prediction task, with a decent extent of generality. In the future, the generalization ability of TSception across subjects will be explored. The effect of segment length in cropped experiments on TSception should also be considered and studied.

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非对称空间层的核。设计了一个高级融合层，以进一步学习 EEG 的半球/全局表示，并减小模型尺寸。

为了获得我们方法的泛化评估，我们在两个基准数据集上对裁剪后的试验进行试验级交叉验证。正如第 4.4 节所述，如果在裁剪实验中将不同试验的样本随机打乱，然后再将数据分为训练集和测试集，那么可以得到非常高的分类结果，但当试验中高度相关的相邻片段没有被模型看到时，这些结果会下降[38], [51]。因此，采用试验级 10 折交叉验证，以确保每个试验中高度相关的相邻片段不会同时出现在训练集和测试集中。为了进一步将我们的方法与现有文献中使用泛化评估设置的方法进行比较，我们进行了留一试验交叉验证，并在每个试验的片段预测上使用投票机制。关于评估指标，我们也遵循[38]，除了准确率外，还添加了 F1 分数，以在非平衡数据集上获得更好的评估。

根据表 3、4 和 5 中两个公开数据集的结果，所提出的 TSception 在大多数实验中实现了比对比方法更高的分类结果。特别是，TSception 的可训练参数只有其对应方法的 1/4 或 1/10。这种效率和效果可能有利于神经网络在现实世界 BCI 应用中的在线使用。广泛的消融研究和可解释性实验表明，TSception 中的所有模块都对分类结果的提升有积极贡献，我们的方法从情绪相关信息中学习。根据表 6，我们发现非对称空间层对分类结果贡献最大。为确保神经网络学习的是情绪相关信息而非无关特征，我们获取了显著性图来可视化神经网络自身识别的最具信息量的区域。图 5a 和 5d 中所有受试者的平均显著性图在前额叶、颞叶、顶叶和枕叶区域显示出强烈激活。然而，图 5b 和 5e 中 F1 分数高的受试者的显著性图仅在额叶、颞叶和顶叶区域显示出强烈激活，这与[1]、[44]、[46]、[47]、[49]、[50]、[52]一致。在 F1 分数最高的前 10%受试者的平均显著性图（图 5e）中，也观察到右侧大脑半球偏侧化模式，这表明右侧大脑半球对效价识别更具信息量。神经科学研究表明[53][54]，大脑的右半球在情绪处理中具有特殊作用。然而，如图 5b 所示，对于高 F1 受试者来说，右半球偏侧化并不存在于唤醒过程中。这可能是因为额叶区域提供的信息足以让神经网络做出决策。此外，我们发现所有受试者的枕叶活动也对神经网络的推理过程有贡献，如图 5a 和 5d 所示。高枕叶活动的一个可能原因是 DEAP 中使用了音乐视频作为刺激。然而，枕叶活动提供的信息对于高 F1 受试者（无论是唤醒还是效价）都较少有用。这表明，在情绪识别方面，枕叶不如其他脑区（如额叶和颞叶区域）信息量丰富。

总之，我们提出了一种多尺度卷积神经网络，名为 TSception，用于捕获脑电图（EEG）情绪识别中的时间动态和空间不对称性。通过使用广义交叉验证策略，所提出的方法和几种基线方法在两个公开可用的基准数据集上进行了评估。所提出的方法在唤醒度-效价预测任务上表现出良好的性能，具有相当程度的泛化能力。未来，将探索 TSception 在跨被试者上的泛化能力。在裁剪实验中，TSception 受片段长度的影响也应被考虑和研究。

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Yi Ding (Student Member, IEEE) received the bachelor's degree from the School of Information Science and Technology, Donghua University, Shanghai, China, in 2017, the master's degree from the School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore, in 2018. He is currently working toward the PhD degree with the School of Computer Science and Engineering, Nanyang Technological University, Singapore. His research interests include brain-computer interface, deep learning, neural signal processing, affective computing and artificial intelligence.



Neethu Robinson (Member, IEEE) received the BTech degree in electronics and communication engineering from the University of Kerala, India, in 2010, and the PhD degree in computer engineering from Nanyang Technological University, Singapore, in 2015. She is currently a senior research fellow and program manager with the School of Computer Science and Engineering, Nanyang Technological University, Singapore. From January to March 2012, she was a visiting researcher with the research team with the Institute of Medical Psychology and Clinical Neurobiology, University of Tübingen, Germany. She served as assistant professor with the Department of Electrical, Electronics and Instrumentation Engineering, Birla Institute of Technology and Science, Pilani, K.K. Birla Goa Campus, India from June 2015 to September 2016. Her research interests include brain-computer interfaces, neural signal processing and machine learning for brain-computer interfaces.

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e1013-e1020 页, 2020 年。



丁怡 (IEEE 学生会员) 于 2017 年获得中国上海东华大学信息科学与技术学院的本科学位, 于 2018 年获得新加坡南洋理工大学电气与电子工程学院的硕士学位。目前, 他正在新加坡南洋理工大学计算机科学与工程学院攻读博士学位。他的研究兴趣包括脑机接口、深度学习、神经信号处理、情感计算和人工智能。



Neethu Robinson (IEEE 会员) 于 2010 年在印度喀拉拉大学获得电子与通信工程学士学位。2013 年在新加坡南洋理工大学获得计算机工程博士学位。她目前是新加坡南洋理工大学计算机科学与工程学院的一名高级研究学者和项目主管。2012 年 1 月至 3 月, 她作为访问研究员加入德国图宾根大学医学心理学与临床神经生物学研究所的研究团队。2015 年 6 月至 2016 年 9 月, 她在印度比拉理工学院皮兰尼 K.K. 比拉果阿校区电气、电子与仪器工程系担任助理教授。她的研究兴趣包括脑机接口、神经信号处理以及脑机接口的机器学习。



Su Zhang received the bachelor's degree from Xiamen University, China, in 2013, and the master's degree from Yunnan Normal University, in 2018. He is currently working toward the PhD degree with the School of Computer Science and Engineering, Nanyang Technological University, Singapore. His current research interests include machine learning, deep learning and emotion recognition.



Qiu Hao Zeng received the bachelor's degree from the School of Astronautics, Harbin Institute of Technology, and the master's degree from the Department of Electrical and Computer Engineering, National University of Singapore. His main research interests include machine learning and brain-computer interface, transfer learning. He is currently a research associate with the School of Computer Science and Engineering, Nanyang Technological University, Singapore.



Cuntai Guan (Fellow, IEEE) received the PhD degree from Southeast University, China, in 1993. He is currently a president's chair professor with the School of Computer Science and Engineering, and director of Artificial Intelligence Research Institute, Nanyang Technological University, Singapore. His current research interests include brain-computer interfaces, machine learning, neural signal and image processing, data analytics, and artificial intelligence. He is a recipient of the Annual BCI Research Award, the IES Prestigious Engineering Achievement Award, the Achiever of the Year (Research) Award, and the Finalist of President Technology Award. He is an associate editor of the *IEEE Transactions on Biomedical Engineering*, *IEEE Transactions on Artificial Intelligence*, *Brain Computer Interfaces*, etc. He is a fellow of AIMBE.

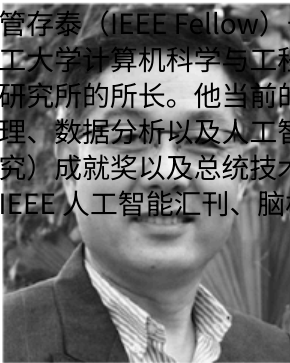
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苏张于 2013 年获得厦门大学的学士学位，2018 年获得云南师范大学的硕士学位。目前，他正在南洋理工大学计算机科学与工程学院攻读博士学位。他的当前研究兴趣包括机器学习、深度学习和情感识别。



曾秋浩获得哈尔滨工业大学航天学院的学士学位，以及新加坡国立大学电气与计算机工程系的硕士学位。他的主要研究方向包括机器学习、脑机接口和迁移学习。目前，他是新加坡南洋理工大学计算机科学与工程学院的研究助理。



管存泰 (IEEE Fellow) 于 1993 年获得中国东南大学的博士学位。他目前是南洋理工大学计算机科学与工程学院的总统讲席教授，以及新加坡南洋理工大学人工智能研究所的所长。他当前的研究兴趣包括脑机接口、机器学习、神经信号与图像处理、数据分析以及人工智能。他是年度 BCI 研究奖、IES 杰出工程成就奖、年度 (研究) 成就奖以及总统技术奖决赛候选人的获得者。他是 IEEE 生物医学工程汇刊、IEEE 人工智能汇刊、脑机接口等期刊的副主编。他是 AIMBE 的会士。

" 如需了解更多关于此主题或其他任何计算机相关主题的信息，请访问我们的数字图书馆 www.computer.org/csdl。